

Contest Segmentation, Recognition, and the Youth Talent Pipeline

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June 2026

Abstract

Schools and youth programs sort participants into ability groups and recognize top performers within each. Tracking bundles peers, instruction, and local rank; this paper isolates the rank-and-recognition channel on the margin compulsory schooling hides—whether participants come back at all. I use administrative panels from Utah high-school mountain-bike racing (2018–2025) and five other state leagues, where ability-group assignment leaves riders of similar speed facing different odds of standing on a podium. Instrumenting competitive success with the recognition a rider’s assigned field makes attainable, each additional top-five finish raises next-season participation by about five percentage points in the primary design, with estimates consistent with a common effect across leagues. What differs across leagues is the supply of recognition structure creates: subdividing fields roughly doubles the top-five finishes awarded—a recognition-supply policy worth up to a point of annual persistence. Because early performance only weakly predicts eventual success, retaining marginal participants has option value.

Keywords: Tracking, ability grouping, contest design, ordinal rank, recognition, persistence, youth sports, instrumental variables, microsimulation

JEL Classification: I21, I28, J24, D82

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1 Introduction

How should an organizer partition heterogeneous participants into ability groups, and how deeply should recognition reach within each? Every tracked classroom, gifted program, leveled course sequence, and youth league answers these questions, almost always by convention rather than by evidence. Two literatures bear directly on the answer and stop just short of it. The economics of tracking asks whether ability grouping raises *achievement* (Betts, 2011; Duflo, Dupas and Kremer, 2011; Card and Giuliano, 2016; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu, Angrist and Pathak, 2014), but the treatment it studies is a bundle: assignment changes a participant’s peers, the instruction aimed at them, and their *local rank*—the position they hold among the people they are compared with every day. The rank-effects literature has since shown that local rank, holding ability fixed, changes investment and later outcomes (Murphy and Weinhardt, 2020; Elsner and Isphording, 2017; Elsner, Isphording and Zölitz, 2021; Denning, Murphy and Weinhardt, 2023)—but it takes the partition as given. Neither literature treats the partition itself as the choice variable, and both study settings that hide the margin that matters most where participation is elective: not what a participant achieves, but whether they come back at all. A fifth grader cannot drop out of fifth grade; a student in an elective math circle, a music program, a coding club, or a sports league can simply not re-enroll, and their continuation—not contemporaneous performance—is the outcome that determines who is still in the pipeline years later.

This paper isolates the rank-and-recognition channel of tracking in a setting built for the purpose. In interscholastic mountain-bike racing, every participant rides the same course on the same day, so the instruction-targeting channel of tracking is switched off by construction; peer quality is perfectly measured (in seconds) and can be controlled or instrumented; and the league re-enrolls riders every fall, so the participation margin is a real, repeated, observable choice for every participant. The league segments riders within grade and gender into ability divisions and publicly recognizes the top five finishers in each division at each race. Because assignment to divisions is governed by a written placement table that is far from

a deterministic ability sort—roughly half of assignment is unexplained by lagged ability—riders of nearly identical speed routinely face different local competition and correspondingly different odds of standing on a podium. The question is whether that manufactured local success causes riders to stay, and, if it does, what the league (or, by careful analogy, a school) should do with the levers it controls: the number and size of ability groups and the depth of recognition.

This setting is the cleanest available laboratory for a question education data structurally cannot answer, and it offers four things at once. First, the participation margin is a genuine annual choice observed for everyone, not a censored or compulsory one. Second, performance is measured in seconds rather than test scores, so ability and peer quality are observed essentially without error. Third, the assignment rule is written down, replicable, and leaves documented slack, which is the raw material for identification. Fourth, the design levers studied here—how many groups, how big, how deep the recognition—are levers the organizer actually adjusts season to season, so the counterfactuals are policies rather than thought experiments. The findings speak directly to the large class of elective educational activities in which dropout, not achievement, is the first-order margin (Stevenson, 2010; Barron, Ewing and Waddell, 2000).

I proceed in two linked steps. The first is causal: does locally generated competitive success affect whether athletes continue? I use three complementary designs, all leveraging the same primitive—a rider’s expected success depends on the field they are assigned to, which is only loosely tied to their own ability—and I present them in order of the cleanliness of their exclusion restrictions rather than their precision. The first instruments recognition with the strength of *external-school* competitors in a rider’s division, variation driven by other coaches’ choices. The second simulates each rider-season’s expected recognition from pre-season speed and the actual composition of the field, using all the assignment slack at once. The third exploits the sawtooth in predicted local position created by the placement table’s division-size cutoffs, in the spirit of Angrist and Lavy (1999); because the written

rule reproduces only 57 percent of actual assignments, I present this cascade as an honest bounding design that rules out large effects but cannot on its own confirm the others—a failure that is itself informative, since it documents the placement discretion the first two designs exploit. Across designs the per-finish effect on next-season participation ranges from +1.4 percentage points in the cleanest-exclusion design to +5.0 in the most precise one (Section 7). These are not interchangeable estimates of one number; they identify effects for different populations, and on the returning-rider subsample where all three apply the two informative designs agree closely. The spread in the full sample is concentrated among first-year riders, and I discuss it openly rather than averaging it away.

The second step is design evaluation, and it requires taking a stand—or openly declining to—on *what* carries the effect: the discrete recognition events (medals, podiums) or the continuous experience of local standing (where you finish relative to the field, race after race). The two imply different policy conclusions: recognition counts can be manufactured by the designer (more groups and deeper podiums mean more medals), whereas local standing is nearly zero-sum (someone must be beaten for someone to beat). I run the tests that could separate the mechanisms and report them all (Section 7). A horse race that instruments both treatments at once favors counts: the top-five count carries the effect while continuous standing, conditional on counts, is if anything negative. A pooled regression discontinuity at the fifth/sixth-place cutoff finds a precise zero at the single-race medal moment, so whatever matters is cumulative over the season rather than a medal ceremony per se. And the per-finish effect declines as the share of the field recognized rises—a dilution pattern with a direct grade-inflation analogue (Butcher, McEwan and Weerapana, 2014). The horse races favor counts, but the dilution pattern sits more naturally with standing-type stories, and no simple version of either mechanism predicts the negative conditional-standing coefficients; I therefore remain agnostic about the mechanism and evaluate every counterfactual under both a recognition-count mapping and a standing-based mapping, reporting both side by side. Recognition-count is the more robust treatment in the current estimates, but the

paper’s design conclusions are stated only where the mappings agree or with the divergence displayed.

The design question has force because of an option-value fact: early performance only weakly predicts who eventually populates the top track. About one-third of the riders who race varsity as seniors came from outside the top three deciles of early performance, and one in seven came from the bottom half. A long-run reduced form using only a rider’s *first race*—before any season-level feedback could operate—shows that early “rank luck” raises eventual varsity participation ($t = 7.1$), and a tier-split version shows the effect is present for non-elite entrants on the varsity margin but absent on the top-decile-ability margin: early success retains and advances ordinary riders into the top participation tier; it does not manufacture elite ability. Retention, in other words, has option value because the league cannot tell at entry who will be varsity-caliber at graduation.

The counterfactuals then quantify the design levers. Holding each season’s riders fixed, recognition-broadening policies (smaller fields, deeper podiums) raise the predicted return rate by up to 0.8 percentage points under the recognition-count mapping, while a pure standing mapping implies near-zero aggregate effects of any lever—precisely because standing is zero-sum, so design can redistribute it but not create it. A dynamic microsimulation that follows real entering cohorts forward shows how modest annual gains compound: under the count-based mapping, top-ten recognition raises the share of entrants who reach their senior year from 30.4 to 41.8 percent and varsity-level production by roughly a third, with about nine-tenths of the gain flowing through participation rather than ability. A bridging decomposition traces exactly where the amplification comes from (survival compounding within the simulation’s endogenous fields, not ability evolution), and all dynamic conclusions are explicitly conditional on the count mechanism being the operative one.

Utah is the centerpiece, but the paper’s claims are about tracking and persistence, not about one state. I assemble parallel administrative panels for five additional interscholastic leagues—Colorado, Minnesota, Wisconsin, Virginia, and Alabama, 47,000 panel rider-

seasons beyond Utah’s 29,000—and run each through the same entity resolution, censoring rules, and estimators. Three portable facts result. The persistence-by-rank gradient appears in every league, including untracked Colorado, which is why descriptive gradients cannot carry causal weight. The causal price of a top-five finish is positive and not statistically distinguishable across leagues (pooled, +2.7 percentage points per finish; excluding Utah, +2.3 and not significant)—precisely measured in Utah, the largest panel with the strongest first stages and the only league examined whose divisions are assigned by administrative rule rather than chosen, and individually underpowered elsewhere. What differs across leagues is the *supply* of recognition that structure creates relative to each league’s own merged-field benchmark: Utah’s divisions award 2.35 times the top-five finishes that merged fields would, Minnesota’s and Wisconsin’s partial structures 1.9–2.2 times, Colorado’s undivided fields 1.04 times. At a roughly constant price, leagues that subdivide mint more of what keeps riders racing—as a partial-equilibrium accounting, worth roughly half a percentage point to 1.2 points of aggregate annual persistence in the divisioned leagues, and approximately nothing in Colorado.

The paper contributes to three literatures. To the economics of tracking and ability grouping (Betts, 2011; Duflo, Dupas and Kremer, 2011; Card and Giuliano, 2016; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu, Angrist and Pathak, 2014), it offers the cleanest available estimate of the local-rank-and-recognition channel, on a participation margin that school settings cannot observe, together with actual design counterfactuals—how many groups, how deep the honors—rather than a binary tracked/untracked comparison. To the ordinal-rank literature (Murphy and Weinhardt, 2020; Elsner and Isphording, 2017; Elsner, Isphording and Zölitz, 2021; Denning, Murphy and Weinhardt, 2023), which established that rank affects investment and outcomes conditional on a given partition, it adds the planner’s problem: if rank matters, the partition itself is a policy instrument, and I estimate and simulate its effects. To contest design and the economics of awards (Lazear and Rosen, 1981; Moldovanu and Sela, 2001; Frey and Gallus, 2017; Tran and Zeckhauser, 2012), it contributes

field evidence on how prize structure and segmentation affect a dynamic behavioral margin—continued participation—rather than contemporaneous effort, including direct evidence on recognition dilution.

The paper proceeds as follows. Section 2 describes the institutions, including what recognition consists of and how the placement table works. Section 3 sets out a minimal dynamic framework that defines option value and the planner’s problem. Section 4 describes data, linkage, and measurement. Section 5 establishes the descriptive dynamics and the option-value facts. Section 6 presents the three identification strategies and their validation. Section 7 reports causal estimates and the mechanism evidence. Section 8 runs the same designs in five additional leagues to test portability and to quantify how much recognition each league’s structure creates relative to its own merged-field benchmark. Sections 9 and 10 evaluate counterfactual designs annually and dynamically. Section 11 concludes. A descriptive comparison to an untracked sport (alpine ski racing) appears in Appendix F.

2 Setting and Institutions

2.1 The league

The Utah High School Cycling League (UHSCL), a member of the National Interscholastic Cycling Association (NICA), grew from roughly 500 riders at its founding to over 3,500 by 2024, making it one of the largest youth cycling programs in the country. Races are held on fall weekends: four to five regional races followed by a State Championship. Utah is divided into geographic regions (four in 2018–2021, six from 2022), and riders compete in the region of their school. I exclude State Championship events throughout, which pool regions and use a different bracket structure—consistent with the league’s own placement formula, which also excludes the championship. Riders re-register each fall; there is no obligation to return, and roughly a third of riders in a typical season do not. That re-enrollment choice is the central outcome of this paper.

2.2 What recognition consists of

Education readers should picture the analogue of honor-roll assemblies rather than professional podiums. At each regional race, the top five finishers in each division are called forward at the post-race awards ceremony, in front of the assembled teams and families; each receives a medal, and their names and teams are announced. Results are posted publicly and accumulate into season series standings, which carry their own end-of-season recognition. For a teenager, a top-five finish is a visible, named, repeated public event witnessed by their school community—arguably closer in social content to making the honor roll than to winning a professional race. The fifth-place rider in a division of forty receives this recognition; the sixth-place rider, seconds behind, does not. Recognition is therefore discrete, local (defined within the division, not the league), and designable: its depth (top five) and its scope (the number and size of divisions) are league choices.

2.3 Categories, divisions, and the placement table

Riders are classified into levels (Middle School, Freshman, Sophomore, JV, Varsity) and, within the Freshman and JV levels, into ability divisions (A through E, strongest to weakest). The Category Placement Table (CPT) assigns riders *within region and year* by, in order: previous-year category, overall series points, and average lap-time rank over their best three races. The league sets target division sizes based on projected region size, then fills divisions in CPT-rank order until each target is met.

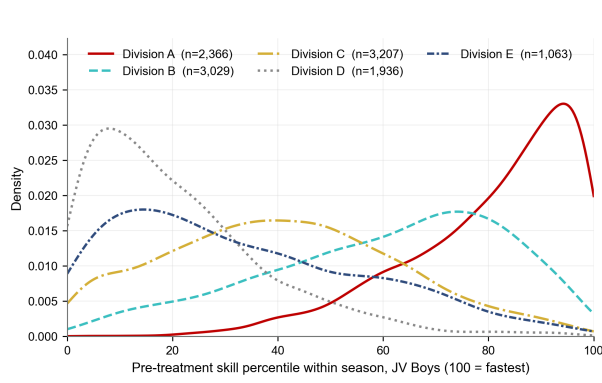
A worked example makes the cascade concrete. Suppose Region 2 JV Boys has 160 riders and the league’s target division size is 40. The CPT first orders the 160: returning JV A riders ahead of returning JV B riders, and so on by previous category; within a previous category, by last season’s series points; among riders without points (or to break ties), by lap-time rank. The first 40 riders in this order become JV A, riders 41–80 become JV B, and so on. The rider ranked 40th and the rider ranked 41st may be separated by a handful of series points—or by nothing more than a tie-break—yet one will spend the season as the

weakest rider in a strong field and the other as the strongest rider in a weaker field, with correspondingly different odds of ever being called to a podium. The cutoffs at 40, 80, and 120 are the “Maimonides” discontinuities exploited by the third research design (Angrist and Lavy, 1999).

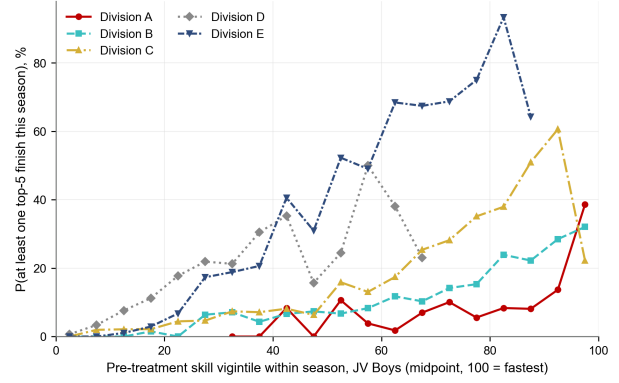
Two further institutional features matter. First, the ranking is a strong *ratchet*: returning Varsity riders stay Varsity, and downgrades are granted only for medical reasons or into the non-competitive Single-Lap category, so riders effectively cannot be placed below their prior category. Second, riders may petition to change category, but only through the third race of the season; within-season division changes are rare, and I exclude post-switch observations from the analysis.

2.4 The slack in assignment

The CPT is informed by ability but is far from a deterministic ability sort. Regressing division rank (A=1 through E=5) on lagged speed, year effects, and demographics yields $R^2 \approx 0.47\text{--}0.49$ within the most finely sorted brackets: roughly half of division assignment is unexplained by lagged skill. The residual reflects coach discretion (coaches frequently petition to keep teammates together), team composition, rider preferences, imperfect information about off-season development, and the mechanical interaction of the cascade with region size. Figure 1 shows the consequence. The left panel plots the distribution of within-season skill across JV divisions: the divisions overlap substantially, so riders of similar ability routinely compete in different divisions. The right panel shows what that implies: at the same skill level, riders in different divisions face sharply different probabilities of a top-five finish. This overlap is the identifying variation for everything that follows.



(a) Skill overlap across JV divisions



(b) Recognition probability by skill and division

Figure 1: Identifying variation. Riders of similar ability compete in different divisions (left) and therefore face substantially different recognition probabilities (right).

Notes: Within-season skill percentile among JV Boys. Substantial overlap implies that division assignment is not a strict ability sort.

3 Conceptual Framework

This section sets out a deliberately minimal model of dynamic participation under ability grouping. Its purpose is not to be estimated but to define the objects the empirical sections measure—the recognition and standing responses, option value, the retention–selection frontier—and to clarify why the mechanism question (counts versus standing) is not a detail but *the* design question.

3.1 Riders

A rider i enters season t with ability a_{it} and an idiosyncratic taste for the activity θ_i . During the season the rider is assigned to a local field (division) and experiences two kinds of competitive outcomes: a recognition count m_{it} (the number of races in which they finish in the recognized set, e.g. top five) and a continuous local standing r_{it} (the average fraction of their field they beat). The rider returns next season if the experienced utility of participation exceeds an outside option c_{it} :

$$P_{i,t+1} = \mathbf{1}\{u(m_{it}, r_{it}; \theta_i) + \delta W_{i,t+1} \geq c_{it}\},$$

where u is increasing in both arguments and $W_{i,t+1}$ is the continuation value of remaining in the activity. The empirical Sections 6 and 7 estimate $\partial P/\partial m$ and $\partial P/\partial r$, the causal participation responses to recognition and to standing, using variation in (m, r) generated by quasi-random field assignment rather than by ability.

3.2 Ability evolves noisily; exit is absorbing

Ability follows a noisy autoregressive process, $a_{i,t+1} = \rho a_{it} + \eta_{i,t+1}$ with $\rho < 1$, and a rider’s eventual standing—in particular, whether they are varsity-caliber by their senior year—depends on the whole path of shocks. Exit is effectively absorbing: a rider who does not re-enroll rarely returns. Together these two facts generate *option value*: when early signals are noisy ($\text{Var}(\eta)$ large relative to the cross-sectional spread of a), the probability that a currently unremarkable rider ends up varsity-caliber is bounded well away from zero, so retaining them preserves a real claim on future talent that exit extinguishes. Section 5 measures the noise directly (the transition matrix from early deciles to senior outcomes) and shows the claim is quantitatively large: about a third of eventual varsity seniors were not early standouts.

3.3 The planner and the zero-sum distinction

The league chooses a design $d = (K, k)$: the number (and hence size) of divisions K , and the recognition depth k (podium places per division). The design changes the joint distribution of (m_{it}, r_{it}) across riders. Two properties of this mapping drive everything in Sections 9 and 10:

- **Recognition counts are manufacturable.** Total recognition per race is $K \times k$: more or smaller divisions, or deeper podiums, mechanically create more recognition events. If u responds to m , the planner can raise aggregate persistence.
- **Standing is (nearly) zero-sum.** The population distribution of fraction-beaten is

essentially fixed—someone must be beaten for someone to beat—so no design can raise *average* standing. If u responds only to r , design choices redistribute persistence across ability levels but cannot create it in aggregate.

This is why the paper treats the mechanism question with such care and why, remaining agnostic, it evaluates every policy under both mappings (Section 9): under a count mechanism the recognition-depth lever is real; under a pure standing mechanism it is inert by construction, and only the distributional consequences of regrouping remain.

3.4 The retention–selection frontier and the planner’s objective

Any design induces a pair: aggregate retention $R(d)$ and the mean ability of the retained pool $\bar{A}(d)$. Designs that protect weak performers from low local standing (small groups, broad recognition) raise R and, mechanically, lower $\bar{A}$ —the retention–selection frontier. A planner who cares about the long-run supply of top-track athletes maximizes neither R nor $\bar{A}$ but the flow of riders who survive to their senior year *and* clear the varsity bar, an object that integrates the participation response, the noise in ability, and the absorbing exit. The annual counterfactuals trace the frontier holding the pool fixed; the dynamic simulation computes the planner’s objective over full careers. When early signals are noisy, the frontier logic favors breadth: the ability cost of retaining marginal riders is low because marginal riders are, with meaningful probability, future varsity athletes. The empirical content of that claim is the transition matrix of Section 5, not an assumption.

4 Data and Measurement

4.1 Sources and linkage

I parse administrative race results for all UHSCL events from 2018 through 2025 from the league’s posted result sheets: 148,876 race-result rows in total. Riders are linked across

races and years with a multi-stage entity-resolution procedure (Appendix A) that infers each rider’s grade from bracket and team information, computes an expected graduation year, and splits same-name riders using graduation-year conflicts, school differences, and career-timeline gaps. The procedure resolves 17,729 riders.

Because “did not race next season” is itself a linkage outcome—a missed match is observationally a dropout—measurement quality is not a technicality here, and I audit it directly (Appendix A). Three results summarize the audit. First, the resolved identifiers contain zero graduation-year conflicts, and only 2.1 percent of riders are associated with more than two schools. Second, among riders who appear to drop out before their senior year, I search for “near-miss” records (same school and gender, similar name) that could represent failed matches; at most 1.2 percent of apparent dropouts have such a candidate, bounding the scope for false dropouts. Third, the headline return rate is nearly invariant to the linkage regime: 71.9 percent under loose (name-and-gender) matching, 71.7 percent under the baseline procedure, 64.2 percent under strict (name, gender, and school) matching, which mechanically breaks links for school-switchers. Finally, match difficulty is balanced with respect to the identifying variation: riders with common surnames (2.8 percent of the sample) have instrument values indistinguishable from others ($t = 0.1$). Out-of-state moves, injuries, and switches to other sports remain observationally equivalent to quitting; the estimates should be read as effects on continued participation *in this league*.

4.2 Sample construction

Table 1 reconciles every sample in the paper in one place. From the 148,876 raw rows, entity resolution and cleaning yield a panel of 140,490 race-level observations; excluding State Championships and incomplete results leaves 120,996 rows, which collapse to 38,654 rider-seasons. Excluding the 2025 cohort (whose 2026 return is unobserved) leaves 32,187 rider-seasons of 17,221 unique riders. Because seniors mechanically cannot return to a high-school league, grade-12 rider-seasons are excluded from all persistence regressions, leaving

the primary persistence sample of 28,470 rider-seasons. The cascade design requires CPT-eligible division brackets and prior-year inputs, so it applies to 17,172 rider-seasons (14,965 after the grade-12 exclusion)—returning riders, roughly half the sample, a fact that matters for interpreting it. The long-run cohort comprises 9,263 riders with an observable senior year (graduation by 2025).

Table 1: Sample accounting

Step	N
Raw parsed race-result rows (all PDFs)	148,876
Rider-resolved panel (v2 identifiers)	143,186
Clean analysis panel (non-competitive categories dropped)	140,490
excluding State Championship races	125,530
finished with valid time and placement	120,996
Rider-seasons	38,654
excluding 2025 cohort (next season unobserved)	32,187
excluding grade 12 (persistence sample)	28,470
Cascade-eligible (division brackets with CPT pools)	17,172
cascade persistence sample (non-senior)	14,965
Unique riders in analysis sample	17,221
Long-run cohort (observable senior year, graduation ≤ 2025)	9,263

Notes: Cascade rows are subsets of the excluding-2025 sample. The grade-12 exclusion reflects that seniors mechanically cannot return to a high-school league; including them would attenuate every persistence estimate with structural zeros.

4.3 Treatments and outcomes

The primary treatment is the rider’s seasonal recognition count: the number of top-five finishes, expressed throughout on an attendance-standardized basis (the per-race top-five rate multiplied by the median season length of three races), so that the treatment does not mechanically embed how many races a rider chose to attend; the number of races entered is separately controlled everywhere (Section 6 explains why this matters). As a continuous alternative, I use the season-average fraction of the field beaten. The primary outcome is an indicator for any race appearance the following fall. Long-run outcomes include ever racing as a senior, racing *varsity* as a senior (the policy-relevant top-track outcome), and a stricter

top-decile-ability benchmark.

4.4 Pre-treatment ability

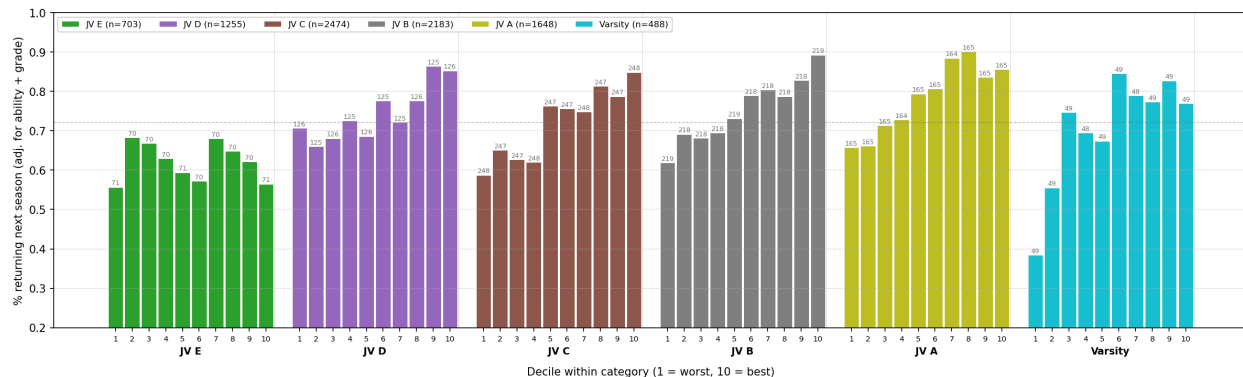
Pre-treatment ability is the course-adjusted speed residual from a regression of log average lap time on venue, year, category, and laps ($R^2 = 0.41$), measured using only prior-season information and shrunk toward grade $\times$ gender $\times$ year cell means by empirical Bayes (Appendix B). The measure is highly stable (season-to-season autocorrelation 0.82), which validates it as a skill measure and simultaneously makes the unexplained half of division assignment more striking. For riders in their first season—who have no prior season—ability and expected-recognition measures used in the long-run analyses are built from the *first race only*, with outcomes defined over subsequent races, so that the right-hand side is predetermined with respect to everything it is asked to explain (Section 6 and Appendix B).

5 Descriptive Dynamics and Option Value

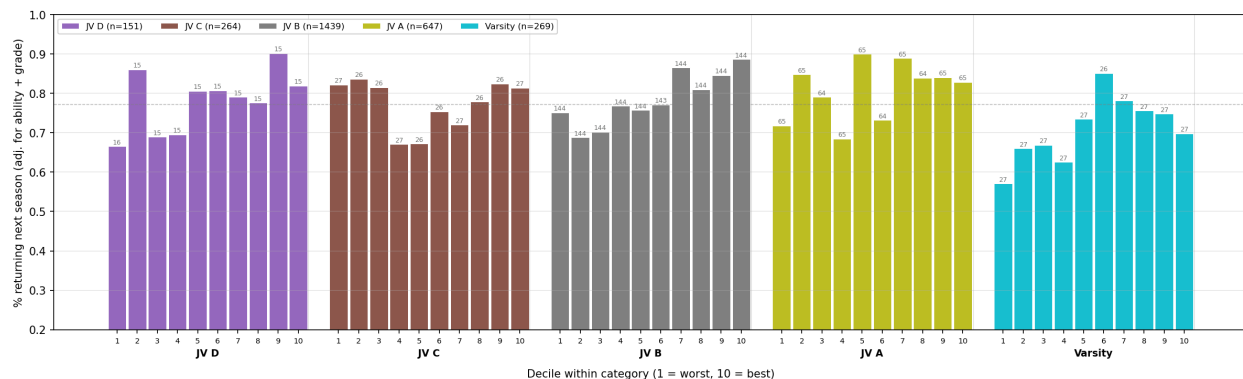
5.1 The motivating fact: local standing tracks persistence within every group

The pattern that motivates the paper is visible in the raw data. Figure 2 plots, separately for each division, the share of riders who return the following season against their finishing decile *within their own division*, after adjusting for ability and grade. Within essentially every division—from JV E to Varsity, for both boys and girls—riders who finish higher relative to their local peers return at higher rates. Because the adjustment nets out a rider’s absolute speed and grade, this is not merely that faster or older riders persist: it is *local* standing, conditional on ability, that tracks continued participation. A rider of given speed placed where that speed ranks well looks more likely to stay than the same rider placed where it ranks poorly. The rest of the paper asks whether that relationship is causal and

what it implies for design.



(a) Boys



(b) Girls

Figure 2: Return rate by within-division performance decile

Notes: Share returning the next season (adjusted for pre-treatment ability and grade) by finishing decile within division (1 = worst, 10 = best), separately by division. The dashed line is the overall adjusted mean. The upward within-division gradient—present for both genders and net of ability—is the descriptive analogue of the paper’s causal result.

A two-way cut sharpens the fact and previews the inference problem. Figure 3 splits riders by global ability quintile and by their local experience: those who collected at least one top-five finish, those who ran in the front of their division without one, and the rest. Celebrated riders return at 84–90 percent across the *entire* ability distribution—a bottom-quintile rider with a podium out-returns a top-quintile rider who finished mid-pack—while uncelebrated mid-pack riders display the ordinary ability gradient (67 to 77 percent). The vertical gap between the celebrated and mid-pack lines is the winner’s boost at fixed ability. But the gap between the celebrated line and the front-runners who never collected a top-

five is far smaller: most of the boost attaches to *reaching the front of a group*, not to the recognition event itself. The figure therefore displays the phenomenon and the identification problem in one place—who reaches the front of a group at a given ability is not random—and the designs in Section 6 are built to isolate the component that the recognition event itself causes.

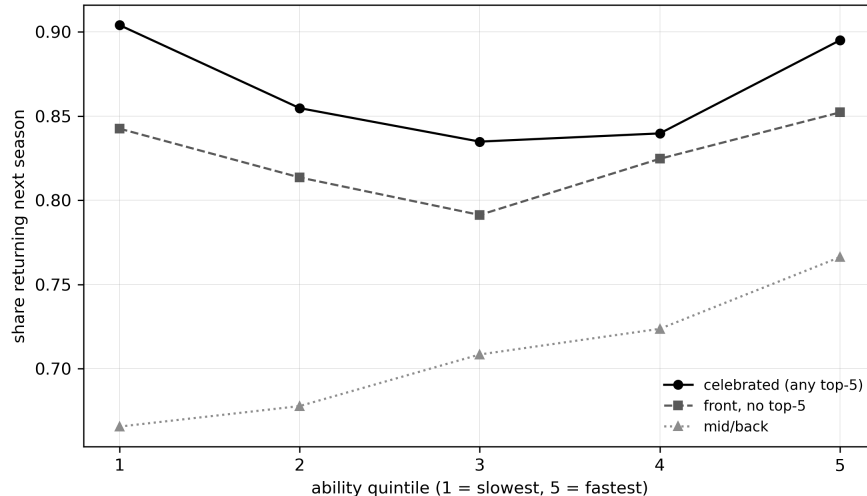


Figure 3: The winner’s boost at fixed ability

Notes: Share returning the next season by global ability quintile (within season×category, pooled across divisions) and local experience in the rider’s own division. “Celebrated” = at least one top-five finish during the season; “front, no top-5” = top three standing deciles within the division without a top-five; “mid/back” = all others. Utah divisioned categories, persistence sample. Cells with fewer than 30 rider-seasons suppressed.

5.2 Early ability badly under-predicts the top track

A second descriptive fact gives the design question its force: early performance imperfectly predicts who eventually reaches the top track. Figure 4 plots the probability of racing varsity as a senior against a rider’s early-years (grades 7–9) performance decile. The gradient is real but far from deterministic: 43 percent of the fastest early decile race varsity as seniors, but so do meaningful shares well down the distribution. Read in reverse, this is the option-value fact: **about one-third of riders who race varsity as seniors came from outside the top three deciles of early performance**, and one in seven came from the bottom half.

Figure 5 fills in the dynamics: the transition matrix from initial decile to senior outcome (panel a) shows mass spread far from the diagonal, and survival in the sport (panel b), while higher for stronger entrants, is substantial across the entire ability distribution. In the language of the framework, the innovation variance in ability is large relative to its cross-sectional spread: the league cannot tell at entry who will be varsity-caliber at graduation, so every early exit destroys a real option.

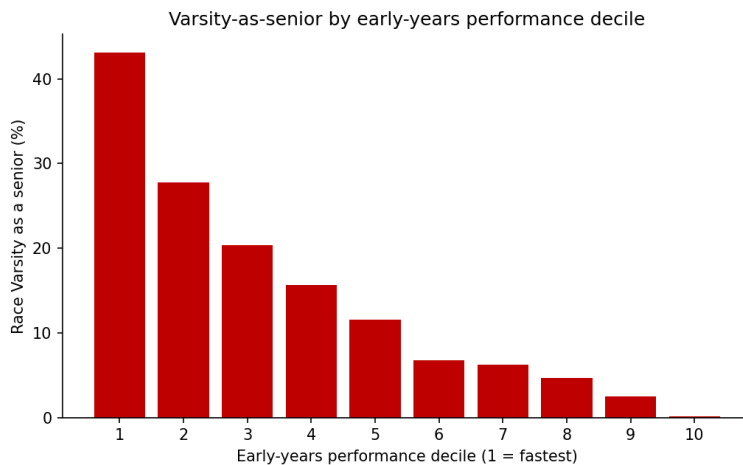


Figure 4: Racing varsity as a senior, by early-years performance decile

Notes: Riders first seen in grades 7–9 with an observable senior year. About one-third of varsity seniors come from outside the top three early-performance deciles; 43 percent of the fastest early decile race varsity as seniors.

5.3 Early rank luck and the long-run pipeline

Does early local success—as opposed to early ability—reshape the pipeline? I regress riders’ long-run outcomes on the expected recognition implied by their *very first race*: the expected per-race top-five probability computed from the rider’s race-one performance and the composition of the field they happened to face, conditioning on initial-ability decile and graduation-cohort fixed effects. Because the measure uses only race one, with outcomes defined afterward, it is predetermined with respect to all subsequent feedback (the timing issue this construction resolves, and the sensitivity of earlier variants to it, are detailed in

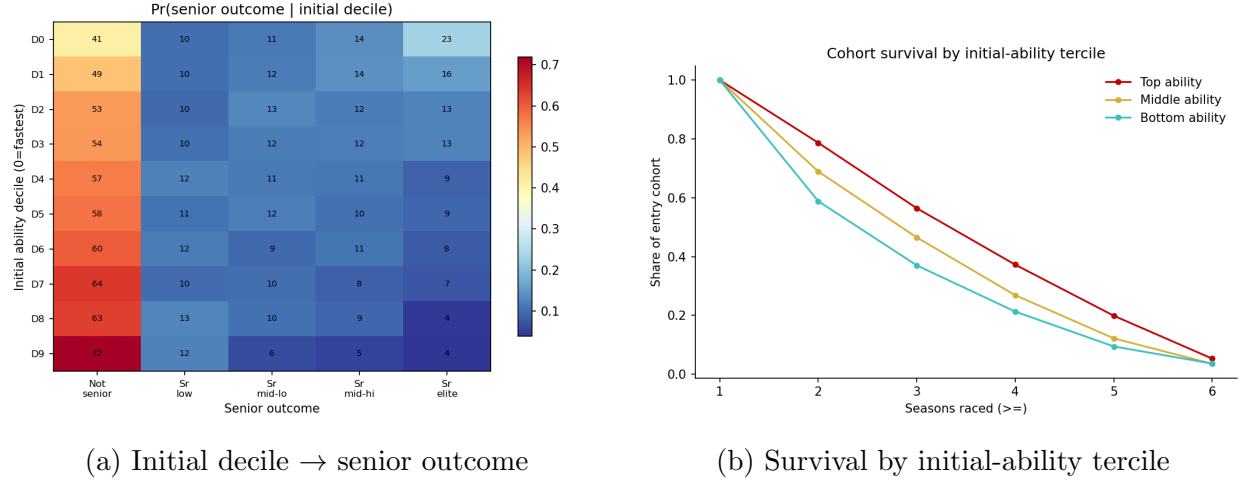


Figure 5: Early ability under-predicts eventual success.

Notes: Cohorts with an observable senior year (graduation ≤ 2025). Initial ability is the first-season pre-treatment skill decile within gender.

Appendix B). Table 2 reports the results. Early rank luck raises eventual *varsity* participation strongly and robustly: +32 percentage points per unit of expected race-one top-five probability ($t = 7.2$), or +15 percentage points per expected top-five count ($t = 8.6$). The estimate is stable under much richer ability controls—+30 with twenty ventile fixed effects and +30 adding a quartic in the race-one performance percentile ($t \approx 6.7$ throughout)—so the result is not an artifact of coarse race-one ability adjustment. Effects on the top-talent and varsity-senior outcomes are similar. By contrast, under this strictest timing the effect on cumulative seasons raced is a null, and the effect on merely racing as a senior is positive but imprecise ($t = 1.6$): what early success appears to change is not the total amount of racing so much as whether a rider stays on the track that leads to varsity.

5.4 Who is moved, and onto which margin

Table 3 splits the race-one reduced form by the rider's initial-ability tier and reports effects *per standard deviation of the instrument within the tier*, alongside each tier's instrument distribution and baseline outcome—the presentation a skeptical reader needs, because the instrument's mass differs enormously across tiers (mean expected top-five count 0.59 for elite

Table 2: Long-run reduced forms: race-one rank luck and the pipeline

Outcome	Per unit of race-1 top-5 rate		Per expected top-5 count	
	Coefficient	t	Coefficient	t
Future seasons raced (count)	+0.06	0.5	−0.01	−0.3
Ever races as senior (pp)	+7.3	1.6	+3.1	1.7
Ever races varsity (pp)	+32.0	7.2	+14.8	8.6
Becomes top talent (pp)	+30.7	9.0	+12.2	8.6
Varsity/elite as senior (pp)	+21.4	6.2	+10.1	7.1

Notes: Each cell is from a rider-level regression of the column outcome on first-race expected recognition, conditioning on initial-ability decile and graduation-cohort fixed effects, with region×entry-year fixed effects. Expected recognition uses only race-one information; outcomes are defined over subsequent races. Non-returns are coded zero for senior/varsity outcomes. SEs clustered by rider. $N = 8,458$. Appendix B reports the original (pre-correction) variant for comparison.

entrants versus 0.03 for the bottom 40 percent). Figure 6 shows the corresponding binned reduced forms. Two results organize the design analysis. First, the varsity-margin effect is positive in *every* tier: +9.2 percentage points per SD for elite entrants ($t = 7.7$), +2.4 for middle entrants ($t = 3.0$), and +2.1 for bottom-tier entrants ($t = 1.9$). Second, the top-decile-*ability* effect exists only for elite entrants (+7.2 per SD, $t = 8.8$) and is essentially zero for everyone else. Early local success moves ordinary riders *into the varsity category*; it does not turn them into top-decile athletes. This asymmetry is the bridge between the causal estimates and the policy analysis: the talent the design margin produces is varsity-level participation, not elite ability, and I hold the simulation in Section 10 to exactly that standard.

6 Identification

The correlation between competitive success and persistence conflates ability, unobserved commitment, and the causal effect of success itself. I isolate the last channel with three complementary designs, all built on the same primitive: a rider’s expected competitive success depends on the field they are assigned to, which is only loosely tied to their own ability. I present them in order of the cleanliness of their exclusion restrictions, not the size

Table 3: Race-one rank luck by initial-ability tier: effects per SD of the instrument

Initial tier	Effect per SD of instrument (pp)			Instr. mean (SD)	Base varsity (%)	N
	Ever senior	Ever varsity	Top decile			
Elite (top 20%)	+2.6 (2.1)	+9.2 (7.7)	+7.2 (8.8)	0.60 (0.54)	41.9	1,876
Middle (deciles 3–6)	−0.4 (−0.5)	+2.4 (3.0)	+0.3 (1.7)	0.21 (0.30)	23.9	4,400
Bottom 40%	+0.8 (0.7)	+2.1 (1.9)	+0.0 (1.0)	0.02 (0.08)	24.3	2,182

Notes: Each cell reports the effect of a one-SD (within tier) increase in race-one expected top-five count on the column outcome in percentage points; t -statistics in parentheses (robust SEs). Conditioning as in Table 2. “Base varsity” is the tier’s mean ever-racing-varsity. Winsorizing the instrument at its 99th percentile leaves all conclusions unchanged. The top-decile column elite-tier effect partly reflects mechanical persistence of high initial ability; the relevant contrast is its absence in the other tiers.

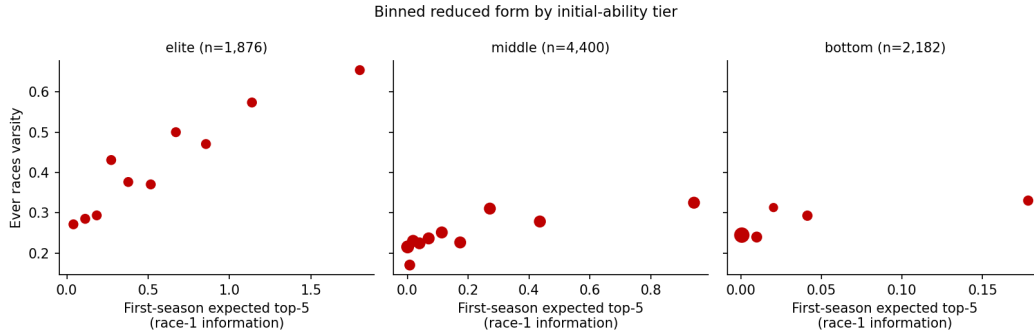


Figure 6: Binned reduced forms by initial-ability tier

Notes: Binned means of long-run outcomes against race-one expected recognition, by initial-ability tier, corresponding to Table 3. The plots show the effects are not driven by thin right-tail leverage within tiers.

of their first-stage F -statistics.

6.1 A pairwise model of expected outcomes

All three designs use a closed-form pairwise win-probability model: rider j beats rider i with probability $\Phi((A_i - A_j)/(\sigma\sqrt{2}))$, where A is pre-treatment ability and σ the race-day residual SD. For any field, the model returns each rider’s expected rank, win, top-five, and fraction-beaten probabilities (tail probabilities via Monte-Carlo ranking so that exactly k riders are top- k in every draw). The model predicts realized placement well out of sample (Spearman 0.74; top-five AUC 0.87; Appendix C and Figure A1). The Monte-Carlo draws use empirical residuals rather than normal deviates, so the fat tail of race-day disasters (mechanicals, crashes) is built into the tail probabilities by construction (Appendix C).

6.2 Design 1: External-opponent instrument

The design I regard as cleanest on exclusion grounds splits each rider’s field into own-school and external-school competitors and instruments recognition with the expected recognition implied by the strength of the *external* competitors only, controlling for own-school strength. Conditional on own ability and own-team environment, the external field a rider faces reflects *other coaches’* placement decisions—which riders from other schools were petitioned up, held back, or newly recruited—and never touches the rider’s own estimated ability. Monotonicity is plausible and testable in sign: a stronger external field weakly lowers every rider’s expected recognition (first-stage coefficients are positive for every subgroup; Appendix Table A6), and I find no subgroup with a reversed first stage. The complier population is broad, though the design moves recognition most strongly for returning riders, whose fields are more predictable.

6.3 Design 2: Simulation instrument

Averaging the pairwise model over the rider’s actual assigned field gives expected recognition for each rider-season—an instrument for realized recognition whose variation, conditional on own ability, comes from the composition of the field the rider was assigned to. This design uses all the assignment slack at once and is by far the most precise. Its vulnerability is that the instrument is a function $g(A_i, \text{field})$ of the rider’s own estimated ability, so misspecification of the ability control $f(A)$ could leak own-ability variation into the instrument. Monotonicity is satisfied by construction: holding own ability fixed, a weaker assigned field raises expected recognition for every rider.

The estimates are stable against exactly this vulnerability. As $f(A)$ is enriched from ability-decile fixed effects to twenty percentile bins to full ability $\times$ grade $\times$ gender interactions, the coefficient barely moves (+5.0, +5.0, +5.4 percentage points; Appendix Table A10). A split-sample construction—ability estimated from odd-numbered races, the instrument built on even-numbered races, eliminating correlated estimation error by construction—yields +4.0 ($t = 6.3$), attenuated as expected under classical measurement error but qualitatively intact.

6.4 Design 3: The cascade (Maimonides) instrument

Reproducing the CPT, I rank riders within region $\times$ year pools by the league’s documented lexicographic rule—previous category first, then prior-season series points (parseable for 97 percent of division-bracket rows), then lagged lap-time ability—and fill the observed division sizes, generating a sawtooth in predicted local position as the CPT rank crosses a division-size cutoff (Angrist and Lavy, 1999). A cubic in the running variable absorbs the smooth ability–rank relationship, so identification comes from the discontinuities alone. Because the CPT requires prior-year inputs, this design applies only to returning riders ($N = 14,958$ in the persistence sample)—roughly half the sample, and a selected half: its compliers are concentrated among returning riders (first stage 0.29, $t = 7.5$) and the fast initial-ability

tercile (0.56, $t = 9.0$), with essentially no first stage among slow-tercile riders (0.08, $t = 2.4$) or girls (0.14, $t = 2.0$; Appendix Table A6). The global first stage is moderate (partial $F \approx 50$) and the running-variable density shows no bunching at cutoffs (Appendix D), but the reduced form is small (-1.3 percentage points), and randomization inference over cutoff placement cannot distinguish it from noise ($p = 0.46$).

A stacked-boundary diagnostic explains why. Restricting to riders within two to eight CPT-rank positions of a division cutoff and stacking all cutoffs with boundary fixed effects and distance trends, crossing the predicted boundary moves neither realized recognition (first-stage t between -0.1 and 1.0 across bandwidths) nor persistence. The reason is institutional: implemented with the league’s own inputs—including actual prior-season series points—the CPT rule reproduces only 57 percent of assignments overall and only 42 percent among riders within two positions of a cutoff, with mispredictions overwhelmingly one division in either direction (Appendix Table A3). Petitions, race-day adjustments through the third race, and coach input dominate exactly where the design needs the rule to bind. I therefore present the cascade as a bounding design: its 95 percent Anderson–Rubin set of $[-16, +4]$ percentage points (donut version, excluding riders within two positions of a boundary: $[-32, +9]$) rules out large effects but does not independently confirm the other designs’ estimates. The failure is informative in its own right: it is direct, rule-based evidence that division assignment retains a large discretionary component conditional on ability and history—the slack that the external and simulation designs exploit. Section 7 returns to what the cascade’s complier population implies for reconciling the designs.

The slot ladder, validated. The placement process can nonetheless be modeled directly, at the cost of two identifying assumptions: take the observed number of *returning* riders in each division at $T+1$ as given, and assume vacated upper-division seats are filled by promoting returners in period- T series-points order (the league ranks on points—a function of placement—rather than race time). This slot-filling model has three testable implications,

and the data confirm all three. First, the no-demotion ratchet binds: only 2.3 percent of within-bracket stayers move down a division. Second, promotions follow points order: among promotion candidates, the rank correlation between period- T points rank and actual promotion is -0.64 and is correctly signed in every cell where it can be computed. Third, divisions do not expand to absorb incumbent surges: regressing upper-division capacity on incumbent supply yields -0.46 rather than the 1.0 of full absorption, so a surge of protected incumbents must crowd marginal riders down the ladder.

Graduation slack and the crowd-out reduced form. The supply of protected incumbents above the B/C margin moves across years for a mechanical demographic reason: how many of last year’s upper-division riders were seniors and graduated (mean senior share 0.15, standard deviation 0.12 within region–bracket pools). Riders who enter the bracket, or sit below the margin, in a high-graduation year face more open seats above them. The reduced form of this graduation slack on the exposed population’s persistence is positive and significant: +87 percentage points per unit of senior share—about +10 percentage points per standard deviation ($t = 2.7$)—with the instrument balanced on exposed riders’ ability ($t = -0.2$); a mild shift toward first-year riders in high-slack years ($t = 1.7$) is controlled throughout (Appendix Table A5).

A promotion-margin experiment. The slot model also produces, for every returning rider below the top division, a predicted promotion cutoff: the number of seats opened above, filled in points order. Comparing riders just inside versus just outside the cutoff (96 pool–year–division cells; 3,669 returner transitions; cell fixed effects and distance trends), crossing the predicted cutoff raises the probability of actual promotion by 11–14 percentage points ($t = 2.0$ – 2.6 across bandwidths), and barely-promoted riders show the expected drop in next-season local standing (-0.03 to -0.05 fraction beaten). The persistence effect at this margin is an honestly bounded zero: +2.3 percentage points (s.e. 8.2) within eight ranks of the cutoff, +5.0 (s.e. 4.7) using all candidates (Table A4). This does not contradict the graduation-

slack reduced form above: the two estimate different objects on different populations —slack moves assignment, peer composition, and cohort thinness together for *all* exposed riders, while the promotion design isolates the single marginal seat for returners—and the promotion interval is wide enough to contain effects of the size the slack reduced form implies. The power arithmetic is unforgiving: with an 11–14 point first stage, detecting the headline +5.0 effect would require resolving a reduced form of less than one percentage point against a standard error near five. The binding constraint is cutoff noise from petitions and multi-division moves, which only the league’s pre-petition placement lists could remove. The ladder design therefore contributes direct institutional validation—the cascading margin exists, and assignment slack is demonstrably discretionary—and a promotion-margin estimate consistent with, but unable to confirm, the other designs (Appendix D).

6.5 Validation: balance, placebos, and the experience confounder

Appendix E collects the validation evidence as numbered tables; here I summarize the design-relevant facts.

The experience confounder. A methodological caution with relevance beyond this paper: instruments built from *continuous* standing (fraction beaten, expected rank) are mechanically confounded by experience. A fast newcomer dominates an easy field, so expected standing is high precisely for riders who are new—and being new independently predicts attrition. The confounding is severe: regressing prior-season participation (a predetermined variable) on the standing-type instrument with only skill and year controls yields $t = -34.8$; adding pre-treatment experience controls (race count, first-year status) collapses it to $t = -1.0$ (Appendix Table A8). All specifications in this paper therefore include experience controls, and any rank-based design in any setting that mixes newcomers with veterans should expect the same issue.

Attendance inside the count. A seasonal recognition *count* summed over attended races embeds attendance—a commitment proxy—inside the treatment: the same logic as the experience confounder, aimed at the count instruments. I therefore standardize: the treatment and instruments are per-race rates scaled by the median three-race season, with races entered controlled everywhere. Appendix Table A9 shows the headline estimate under the old (raw-count) and new (standardized) definitions (+3.7 versus +5.0 for the simulation design; the external design is essentially unchanged, +1.6 versus +1.4), and a fully rate-on-rate specification points the same direction with less precision.

Balance and pre-trends. With experience controls in place, prior-season participation is balanced with respect to the standing instruments ($t = -1.0$) and shows a small residual correlation with the count instruments (coefficient -0.010 , $t = -5.8$)—precisely estimated but an order of magnitude too small to matter relative to the estimated effects, and of the wrong sign to generate them. Lagged *performance* measures correlate with all instruments conditional on ability deciles, as they must (the instruments are functions of lagged ability); this is the $f(A)$ -richness concern addressed above, and the stability of the estimates across much richer $f(A)$ is the relevant test. The full balance matrix, including the regression of season- $(s-1)$ participation on season- s instruments that serves as the pre-trends placebo, is Appendix Table A7.

6.6 Estimating equations and inference

The estimating equations, at the rider-season level, are

$$\begin{aligned}\text{Recognition}_{is} &= \pi Z_{is} + f(A_{is}) + X_{is} + \delta_{rs} + u_{is}, \\ \text{RacedNext}_{is} &= \beta \widehat{\text{Recognition}}_{is} + f(A_{is}) + X_{is} + \delta_{rs} + \varepsilon_{is},\end{aligned}$$

where $f(A)$ is a flexible ability control, X includes pre-treatment experience (races entered, first-year status) and, for the cascade, the running-variable cubic, and δ_{rs} are region $\times$ year fixed effects—the baseline stratification, since divisions are defined within region-years. Because the instruments vary at the field level (division $\times$ region $\times$ season), rider clustering alone risks overstating precision; I report standard errors clustered by rider and, in brackets, by field (436 fields) for every estimate. Two further inference caveats are stated rather than hidden. First, the expected-recognition instruments and the ability control are generated regressors (pairwise model plus empirical-Bayes shrinkage); a full-pipeline bootstrap propagating that estimation error has not yet been implemented, so the reported standard errors do not include generated-regressor uncertainty, and the field-clustered statistics should be treated as the primary guard against overstated precision. Second, for the cascade I supplement asymptotic inference with weak-instrument-robust Anderson–Rubin sets and randomization inference over cutoff placement.

7 Causal Estimates and Mechanism Evidence

7.1 Main estimates

Competitive success raises next-season participation across all three designs, with effects concentrated in the recognition-count treatment (Table 4). Three results stand out.

First, the recognition-count estimates. The simulation design yields +5.0 percentage points per additional top-five finish ($t = 8.0$ by rider; $t = 5.8$ by field)—the paper’s headline estimate. The external-opponent design, whose exclusion restriction I regard as cleanest, yields a smaller +1.4 ($t = 2.0$ by rider; $t = 1.4$ by field). The cascade’s point estimate is an uninformative +1.3 with an Anderson–Rubin set of $[-16, +4]$.

Second, the recognition effect is concentrated among returning riders, and the designs agree there. The cascade identifies effects only for returning riders, and the external design’s recognition variation is also concentrated there. Re-estimating the other designs on

Table 4: Effect of competitive success on next-season participation

Design	OLS	2SLS	(rider SE) [field SE]	First-stage F	N
<i>Panel A. Recognition count (pp per additional top-five finish)</i>					
External opponent	+2.6	+1.4	(0.7) [1.0]	1,987	28,434
Simulation	+2.5	+5.0	(0.6) [0.9]	2,577	28,434
Cascade (CPT)	+3.5	+1.3	(5.0) [4.7]	50	14,958
AR 95% set	[−16, +4] pp; donut [−32, +9] pp; randomization inference $p = 0.46$				
<i>Panel B. Continuous standing (pp per 0→1 fraction of field beaten)</i>					
External opponent	+25.6	+0.5	(2.6) [3.5]	6,530	28,434
Simulation	+22.1	−10.0	(2.0) [3.7]	9,898	28,434
Cascade (CPT)	+22.8	−15.2	(20.2) [21.9]	58	14,958

Notes: Outcome is racing the next season; persistence sample (grade 12 excluded). Panel A treatment: one additional top-five finish, standardized to a median three-race season (attendance-purged; Appendix Table A9). Panel B treatment: moving from beating none to all of one’s field. All specifications include region×year fixed effects, ability-decile fixed effects, and pre-treatment experience controls (races entered, first-year status); the cascade adds a cubic in the CPT running variable. SEs clustered by rider in parentheses and by field (division×region×season; 436 fields) in brackets. The cascade is underpowered; its Anderson–Rubin set and randomization-inference p -value are the appropriate summaries. Reported SEs do not include generated-regressor uncertainty (Section 6).

the cascade-eligible subsample yields +4.4 ($t = 5.0$) for the simulation design and +0.9 for the external design; the cascade’s wide set cannot adjudicate between them. The heterogeneity table below shows both full-sample designs positive for returning riders (+7.4 in the simulation design, +2.5 external), while for first-year riders the estimates are small and design-dependent (+2.2 versus −1.1). Entrant-season estimates lean on the within-season ability measure that Section 6 flags as the weakest input—restricting the simulation’s residual pools to strictly pre-outcome seasons (the construction used throughout) cut the first-year estimates far more than the returning-rider ones—so I treat the returning-rider range of roughly +2 to +7 percentage points as the supportable statement and the entrant margin as unresolved.

Third, the continuous-standing estimates do not survive the count estimates’ scrutiny: precisely estimated but *negative* in the simulation design (−10.0; $t = -5.1$ rider, −2.7 field), a null in the external design, and an imprecise negative in the underpowered cascade. No simple standing story predicts this pattern; it previews the mechanism evidence below.

Table 5 reports the heterogeneity the education audience will ask for first. The count effect is present for both genders (simulation design: boys +5.2, girls +4.4) and is roughly three times as large for returning as for first-year riders in the simulation design (+7.4 versus +2.2)—recognition appears to work on riders with a season behind them, consistent with the returning-rider concentration in the design-overlap discussion above. Fixed-effects and functional-form robustness are collected in Appendix E: the simulation estimate moves from +4.9 (year fixed effects only) to +5.0 (region $\times$ year, the baseline) to +5.0 under school $\times$ year fixed effects, which compare riders on the same team in the same season (Appendix Table A11); enriching $f(A)$ moves it between +5.0 and +5.4 with the split-sample construction at +4.0 (Appendix Table A10).

Table 5: Heterogeneity: gender and experience

Sample	Top-five count (pp per finish)		Fraction beaten (pp, 0 $\rightarrow$ 1)	
	Simulation	External	Simulation	N
Boys	+5.2 (0.8) [1.1]	+0.0 (1.0) [1.2]	−9.3 (2.2) [4.1]	22,097
Girls	+4.4 (1.2) [1.7]	+1.0 (1.3) [1.9]	−10.0 (4.4) [7.0]	6,337
First-year	+2.2 (0.9) [1.4]	−1.1 (0.9) [1.6]	−21.8 (2.9) [6.4]	14,496
Returning	+7.4 (0.9) [1.0]	+2.5 (1.1) [1.2]	−0.5 (3.0) [4.3]	13,938

Notes: 2SLS estimates of the effect on next-season participation, by subsample. Specifications as in Table 4. SEs clustered by rider in parentheses, by field in brackets. Both designs are positive for returning riders; first-year estimates are small and design-dependent; see text.

7.2 Mechanism evidence: counts, standing, or the medal moment?

Whether the effect is carried by discrete recognition events or by continuous local standing determines which design levers work (Section 3), so I run every test available and report them all, committing to none.

The horse race. Table 6 instruments both treatments simultaneously. In the joint 2SLS, the top-five count carries the effect (+10.2 per finish, $t = 12.4$) while fraction-beaten, conditional on counts, is *negative* (−27.6, $t = -10.9$); the joint reduced form gives the same

answer. The less demanding specification that conditions on top-five-count dummies rather than instrumenting jointly agrees in sign (-22.5 per $0 \rightarrow 1$, $t = -8.5$), while the count effect conditional on standing deciles stays positive ($+3.7$, $t = 2.2$). The cascade version is consistent with no count effect and a negative standing effect, but inherits all the cascade’s weakness. Identification in the horse race comes from riders with identical standing but different counts (variation in field size and race-day noise), which is genuine but thinner than the single-treatment variation; and no simple mechanism—count or standing—predicts a conditional-standing coefficient this negative, which counsels caution about over-reading the joint specifications. I read the table as favoring counts, with the standing channel carrying no positive effect anywhere it is measured.

Table 6: Mechanism horse race: recognition count versus continuous standing

Specification	Top-five count (pp)	Fraction beaten (pp)
Joint 2SLS (simulation instruments)	+10.2 (0.8)	−27.6 (2.5)
Joint reduced form	+6.7 (0.5)	−18.9 (1.8)
Top-five fraction-beaten decile FE	+3.7 (1.7)	—
Fraction beaten top-five count FE	—	−22.5 (2.7)
Joint 2SLS (cascade instruments)	+2.5 (5.6)	−22.3 (22.9)

Notes: Outcome is next-season participation; persistence sample ($N = 28,434$; cascade row 14,958). Count effects per additional top-five finish; standing effects per $0 \rightarrow 1$ fraction beaten. Rider-clustered SEs in parentheses. Controls as in Table 4.

The medal moment is not the margin. If recognition operates through the discrete event of medaling at a given race, riders finishing fifth versus sixth should differ in persistence. Pooling every fifth/sixth-place comparison across all races and seasons in a local-polynomial design (Figure 7), the discontinuity is a precise zero: $+0.4$ percentage points (SE 1.0) at a bandwidth of 3.5 placement positions ($N = 15,619$), with a minimum detectable effect at 80 percent power of 2.5 percentage points—well below the per-finish causal estimate. Estimates at narrower bandwidths are similar (-1.4 and $+1.4$, both insignificant). Whatever recognition does, it does not do it through the medal ceremony of a single race: the operative treatment is cumulative over the season. Figure 8 corroborates this nonparametrically:

persistence declines smoothly in season-average placement with no discontinuity at the medal cutoff, and placebo gaps between adjacent finishers are null.

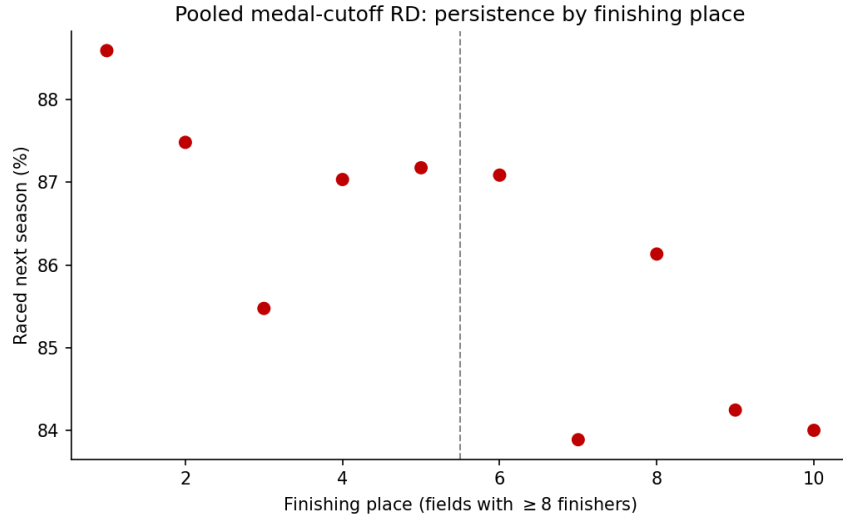
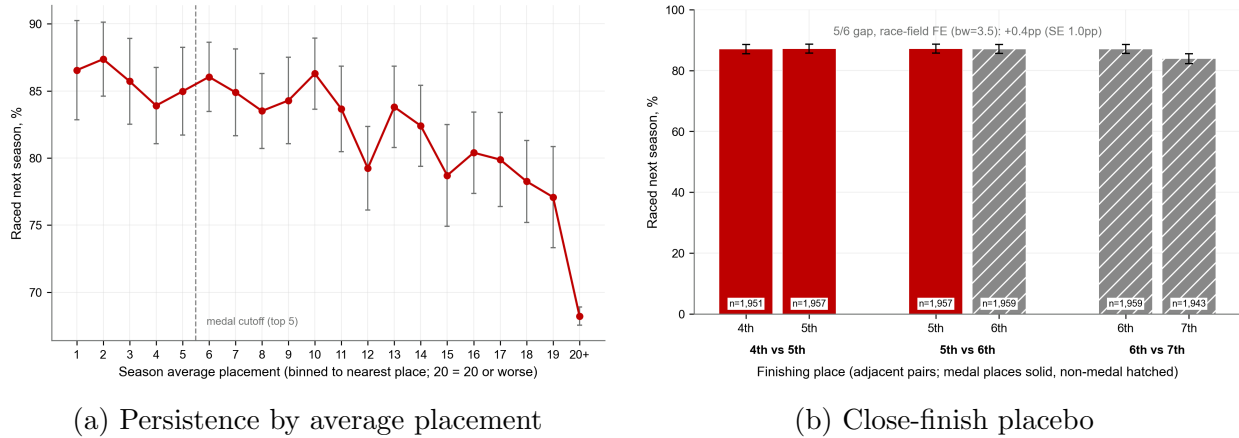


Figure 7: Pooled regression discontinuity at the fifth/sixth-place cutoff

Notes: Next-season participation against single-race finishing position relative to the medal cutoff, pooled across all races and seasons. The discontinuity is +0.4 pp (SE 1.0) at bandwidth 3.5; $MDE_{80} = 2.5$ pp.



(a) Persistence by average placement

(b) Close-finish placebo

Figure 8: The effect is cumulative over the season, not a single-race threshold.

Recognition dilutes. If recognition derives part of its value from scarcity, the per-finish effect should decline where a larger share of the field is recognized—and it does. Interacting the (instrumented) top-five count with the field’s demeaned medal share yields a main effect of +5.0 percentage points and an interaction of -0.19 ($t = -4.1$; Appendix Table A5): each

10-percentage-point increase in the share of the field recognized reduces the per-finish effect by about 1.9 percentage points. This is the athletic analogue of grade inflation eroding the signal value of honors (Butcher, McEwan and Weerapana, 2014), and I carry it into the policy section as a first-order caveat on recognition-broadening counterfactuals, which extrapolate toward exactly the high-share regimes where the effect is weakest.

Counts versus standing as parameterizations. The pairwise model translates one treatment into the other. At observed field sizes, moving from beating none to all of one’s field corresponds to about 1.8 additional top-five finishes per (three-race) season on average (ranging from 1.6 in fields of 40-plus to 3.4 in fields under 15), so if counts were the structural treatment, the +5.0 count estimate would imply a 0→1 standing effect of about +9 percentage points. The directly estimated standing effect is instead −10.0 (Figure 9). The two parameterizations therefore do not describe one dose–response: the count parameterization is stable across designs and controls, while the standing parameterization changes sign depending on what it is conditioned on—which is itself evidence that the discrete recognition event, not the continuous rank, is the treatment riders respond to.

Where the verdict stands. The recognition-count treatment is the more robust object in these data: it survives the joint horse race, both genders, both experience groups, school×year fixed effects, and the attendance purge, while the standing treatment is null or negative in every design and turns significantly negative conditional on counts. But the dilution pattern sits more naturally with scarcity-of-standing stories, the medal-moment null says the operative object is season-cumulative, and no simple version of either mechanism predicts the negative conditional-standing coefficients. The paper’s policy analysis therefore carries both mappings everywhere (Section 9), and readers should hold conclusions that depend on the count mechanism—particularly recognition-depth levers—more loosely than those that survive both.

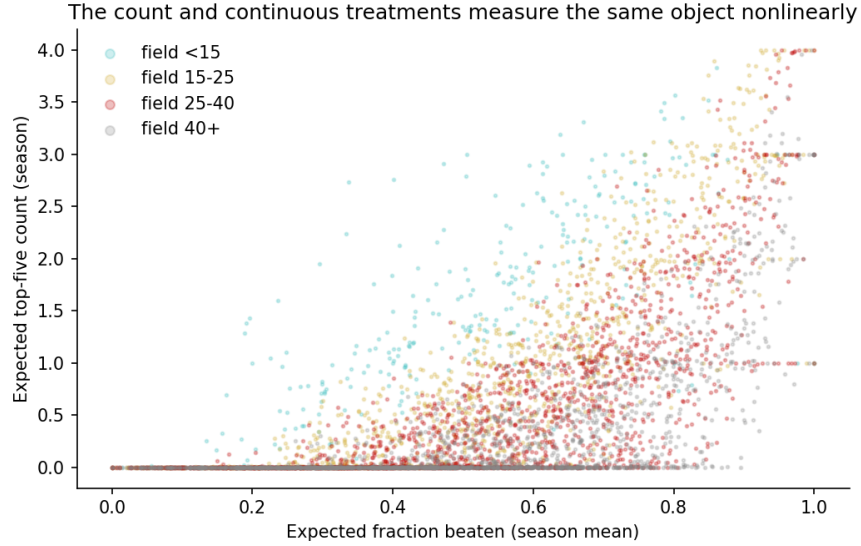


Figure 9: Translating counts into standing through the pairwise model

Notes: Implied 0→1 fraction-beaten effects from the count estimate, by field size, versus the directly estimated standing effect. At typical field sizes a 0→1 standing change corresponds to roughly 1.8 top-five finishes per season.

Participation, not performance. Recognition could in principle operate on an intensive margin as well: riders who taste success might train harder and ride faster. I find no evidence of this, on either of two margins. First, next-season speed: instrumenting competitive success as in the main design, with the same ability-control battery the persistence results survive (Table A10), the effect of an additional top-five finish on next-season course-adjusted lap pace is +0.1 percent (s.e. 0.3)—ruling out speed gains larger than roughly one year of typical adolescent development.¹ Second, contemporaneous effort: the same rider, within the same season, rides no faster in races where the day’s start field places them in podium contention than in races where it does not (within rider×season fixed effects, every expected-rank bin within ± 0.15 percent of the rider’s own average pace), and pace shows no jump at the division-promotion cutoff. The recognition incentive operates on whether riders show up—this season and next—not on how hard they ride once the race starts.

¹Speed outcomes are demanding: pace is nearly collinear with fine-grained ability, so modest ability-control error masquerades as treatment effects (a naive specification yields a spurious +1.6 percent, $t = 5.4$, which the interacted ability controls eliminate), and roughly forty percent of a season-mean pace deviation is transitory and mean-reverts (Table A12).

7.3 The channel decomposition: what is “recognition” made of?

The instruments shift division assignment, which moves a bundle: recognition chances, the division label (A versus E—prestige and stigma, the tracking literature’s classic concern), peer quality, and race difficulty. For an education audience this bundle is not a nuisance but the question: *which* component of track placement moves persistence? Table 7 unbundles it by adding controls that absorb each channel in turn, within the division-bracket subsample where labels exist. The recognition-count effect is +4.4 in that subsample at baseline and is, if anything, larger *within* division labels (+6.6, $t = 6.8$)—so the label itself (being “an A rider” versus “a C rider”) is not what carries the effect. Adding peer-quality controls (division mean and SD ability, own-school teammate count and strength) moves the estimate to +4.5, and adding field size to +3.5 ($t = 3.1$). Recognition chances themselves, not the track’s name or the company it implies, appear to be the operative ingredient—a finding with direct relevance to the question of whether honors-type interventions can be separated from grouping itself (Frey and Gallus, 2017; Tran and Zeckhauser, 2012). The continuous-standing treatment, by contrast, turns sharply negative under the full channel battery (−69, $t = -9.7$)—a magnitude no behavioral story supports, one more indication that the standing parameterization is not the structural object.

8 Evidence Beyond Utah: One Price, Different Supply

The mechanisms studied above are not Utah peculiarities: five additional leagues, processed through the identical pipeline, let me test whether they travel. I assemble administrative race results for five additional interscholastic mountain-bike leagues—Colorado (2012–2025), Minnesota (2015–2025), Wisconsin (2016–2025), Virginia (2016–2025), and Alabama (2021–2026)—and process each through the same pipeline as Utah—the same entity resolution, grade inference and graduation censoring, course-adjusted speed measure, and estimators—so differences in results reflect the leagues, not the methods. Together the six leagues con-

Table 7: Channel decomposition: label, peers, field size

Specification	Coefficient (pp)	t	N
Baseline, full sample	+5.0	8.0	28,434
Baseline, division brackets only	+4.4	4.9	14,931
+ division-label fixed effects	+6.6	6.8	14,931
+ peer-quality controls	+4.5	4.1	14,931
+ field size	+3.5	3.1	14,931
Standing treatment, all channel controls	−68.6	−9.7	14,931
External instrument: baseline (div. brackets)	+0.9	0.8	14,931
+ division-label fixed effects	+2.6	2.4	14,931
+ peer-quality controls	−0.9	−0.7	14,931
+ field size	−4.2	−3.2	14,931

Notes: 2SLS estimates (simulation design) of the top-five-count effect on next-season participation (top panel) and, for comparison, the external instrument (bottom panel), progressively absorbing the non-recognition contents of division assignment. Peer-quality controls: division mean and SD of ability, own-school teammate count and mean strength. Rider-clustered SEs. The final row replaces the count treatment with fraction beaten under the full control set.

tribute 76,300 rider-seasons (panel totals; estimation samples are smaller because the instruments require a prior-season speed measure), 54,300 of them with an observable next-season outcome. The leagues differ in exactly the dimension the paper is about: Utah subdivides each grade-gender category into ability divisions (2.18 separately awarded divisions per category on average); Minnesota, Wisconsin, Virginia, and Alabama subdivide partially (1.2–1.3); Colorado runs each grade-gender category as a single start (1.00). League-by-league return rates range from 63 to 74 percent.

Three facts emerge, and together they discipline how the Utah results generalize.

The persistence gradient is universal—and not causal by itself. The motivating sawtooth of Section 5 replicates in every league and every category block: riders in the bottom decile of their category return at rates 15 to 52 percentage points below riders in the top decile (Figure 12). The descriptive gradient is just as steep in Colorado, where there are no divisions, as in Utah. The gradient alone therefore cannot distinguish skill sorting from a causal effect of competitive success—which is why the paper leans on instruments, and why replicating the *causal* estimate across leagues is the informative exercise.

The per-finish estimate is positive everywhere it is precise, and no league rejects a common price. Table 8 reports, for each league on an identical returning-rider frame, the OLS gradient and the two instrumental-variable designs of Section 6 (the cascade applies only in Utah, below). Pooling all six leagues with league-specific match cells, one additional top-five finish raises next-season participation by 2.7 percentage points ($t = 2.2$). Utah is the only league individually significant (+3.4 to +5.7 across designs), and it largely carries the pooled result: excluding Utah the pooled estimate is +2.3 ($se = 1.5$), positive but not significant. The summary has two halves. Outside Utah the estimates are individually underpowered: the cross-league exercise cannot independently confirm the causal effect, but every league’s confidence interval is consistent with the pooled price. Utah—the largest panel with the strongest first stages, and on this frame matching its own within-paper returning-rider estimate of +3.8 (Table 5)—is where the price is measured.

A direct test of heterogeneity cannot reject equality, but only weakly: interacting the instrumented top-five count with an indicator for division-running leagues yields an increment of +0.5 percentage points ($se = 2.3$), a null that is bounded, not tight—differences in the price smaller than roughly 4.5 percentage points, larger than the pooled price itself, would go undetected. One leg of that comparison is also compromised: Colorado, the only fully untracked league, fails the placebo test below.

The placebo and overidentification evidence bears this out. Weak-instrument-robust inference changes nothing because there was nothing fragile to protect: Anderson–Rubin sets are wide for the small leagues (Virginia, Alabama), and wild-cluster bootstrap p -values for Alabama (0.13) and Virginia (0.14) likewise fail to reject—those leagues are individually uninformative, not confirmatory. Placebo regressions of *prior* speed improvement on the current instrument pass in Utah, Wisconsin, and Minnesota ($p = 0.32, 0.95, 0.23$) but fail in Colorado ($p = 0.003$), so Colorado’s imprecise estimates carry a trajectory caveat. Virginia and Alabama have too few consecutive three-season careers to run the test. Overidentification tests using both instruments jointly reject nowhere ($p = 0.11$ – 0.98). A further cau-

tion applies to single-start leagues: with no divisions, the simulation instrument’s variation comes from race-to-race attendance and entry composition rather than from administrative assignment—a margin closer to rider choice—so the Colorado and Virginia columns are weaker tests of the causal claim than Utah’s.

Table 8: The per-finish effect on an identical frame, six leagues

League	N	OLS Effect of one additional top-five finish on next-season return (pp)	Opponent-strength IV (Design 1)	Simulation IV (Design 2)	Assignment-rule IV (Design 3, cascade)
Utah	8,362	+2.8 (0.4)	+5.7 (2.3) [F=361]	+3.4 (1.9) [F=238]	+4.5 (3.8) [F=131]
Colorado	4,303	+1.8 (0.6)	−5.9 (8.9) [F=23]	+4.3 (3.3) [F=77]	—
Minnesota	3,136	+2.6 (0.7)	+2.9 (4.2) [F=112]	−0.1 (2.8) [F=94]	—
Wisconsin	1,278	+0.6 (1.0)	+0.1 (5.0) [F=60]	+0.2 (2.5) [F=113]	—
Virginia	705	+6.8 (1.9)	weak instr. [F<1]	+11.0 (7.1) [F=33]	—
Alabama	374	+2.2 (2.6)	+5.5 (5.8) [F=42]	+9.0 (6.0) [F=40]	—
Pooled (six leagues)	18,158			+2.7 (1.2)	
Pooled excl. Utah	9,796			+2.3 (1.5)	

Notes: Returning riders with observable outcomes, grade-12 excluded; within-cell demeaning on gender×category×season×skill-ventile cells with a cubic in lagged speed and races entered; standard errors clustered by rider; first-stage F on the row beneath each estimate. “Weak instr.”: point estimates are suppressed wherever the first-stage F is below 10; “—” marks designs that do not exist in a league (the cascade requires a league-administered assignment rule, which only Utah has). N is smaller than the within-Utah returning-rider samples of Section 7 because this frame requires consecutive-season participation for the lagged-speed measure and trims the top and bottom 5 percent of the lagged-speed distribution. Pooled rows stack the six trimmed league samples (so N equals the sum of the league rows) and include league in the match cells. These are returning-rider estimates on a deliberately identical cross-league frame; Utah’s richer within-league designs (Section 7) remain the paper’s primary estimates.

League-administered assignment exists only in Utah. The cascade logic of Section 6 doubles as a structural diagnostic: if a league fills ability divisions by rule, predicted assignment under perfect rank-based sorting should predict actual assignment. In Utah it does (first-stage coefficient 0.50, $t = 37$). In Wisconsin and Minnesota it does not (coefficients 0.05 and −0.02, $t < 1$): their JV2/JV3 “tiers” are race-distance choices made by riders and

coaches, not league-imposed ability assignments. (Virginia and Alabama subdivide too little to run the diagnostic; the claim is about the leagues examined.) Utah is thus not merely the biggest sample; it is the only league examined whose division structure is an administratively assigned treatment—the variation a study of tracking needs, and the reason it anchors the paper. Two clarifications keep this paragraph consistent with Section 6: the 0.50 coefficient is the assignment-on-predicted-assignment first stage, not the recognition first stage (the corresponding instrument F for the top-five count is 131 in Table 8); and it coexists with the earlier finding that the written rule reproduces only 57 percent of individual assignments—the rule predicts the broad division well and the marginal slot poorly, which is exactly why the cascade bounds rather than pins down the effect.

What structure does is change the supply of recognition. If the per-finish price is roughly constant, division structure matters through arithmetic: subdividing a field of fifty into two fields of twenty-five doubles the number of top-five finishes awarded. Table 9 quantifies this with a within-league accounting counterfactual: merge each race’s category-gender field into a single start, hold attendance fixed, and count top-five slots. Two different objects must not be conflated here. The *level* of recognition per capita is dominated by league scale and field size, not structure—small-field Virginia awards more top-fives per 100 rider-seasons (57) than Utah (36), and Colorado (32) is within four of Utah. The structural object is each league’s own *multiplier*: actual slots relative to its own merged counterfactual. On that margin Utah’s divisions multiply recognition 2.35-fold (36 top-five finishes per 100 rider-seasons against 15 merged); Minnesota’s and Wisconsin’s partial structures multiply it 1.9–2.2-fold; Colorado’s single starts multiply it essentially not at all (1.04; the residual 4 percent comes from occasional split and combined-grade starts). Pricing the extra slots at the pooled estimates implies that division structure contributes roughly 0.5 to 1.2 percentage points of aggregate annual persistence in the divisioned leagues and approximately nothing in Colorado, relative to each league’s own merged benchmark. Figure 10 summarizes the

section in one picture: no league’s price is distinguishable from the pooled price; the quantity is a policy choice. The incidence follows from the same arithmetic, and Figure 11 shows it directly: the additional slots accrue to mid-ability riders. A median-ability Utah rider finishes top-five in 42 percent of starts in the lowest division and essentially never in the top division (0 percent); the same median-ability rider in Colorado’s merged starts finishes top-five in under 1 percent of starts. This is the supply-side counterpart of the within-Utah evidence that recognition moves marginal rather than elite riders.

Three caveats bound this accounting. First, it is partial-equilibrium: it holds the price fixed as the quantity changes. The split-sample test of the price in large versus small fields is noisy in both directions and bounds little. More pointedly, the paper’s own dilution estimate (Section 7.2, Table A5) implies the per-finish price *rises* as recognition becomes scarcer, so the merged counterfactual’s price would likely exceed the divisioned price and the constant-price contributions in Table 9 overstate the structural gain; under the dilution-adjusted pricing used for the Utah counterfactuals in Section 9, the implied contributions shrink by roughly half. Second, the non-Utah contributions price slots at the pooled estimate, which Utah largely carries; the excluded-Utah estimate alone cannot confirm a positive price. Third, the price is identified from returning riders—the population for which all leagues support the same design—so the aggregate contribution inherits that LATE interpretation.

A companion comparison to an untracked *sport*—Utah alpine ski racing, which groups athletes only by age—appears in Appendix F; it does not deliver a clean tracked-versus-untracked contrast and is reported there as descriptive context with its confounds stated.

9 Annual Policy Counterfactuals

I now ask how next-year persistence and the retained talent pool would differ under alternative designs, holding each season’s rider pool fixed. For each policy I re-assign riders to divisions, recompute every rider’s expected recognition and standing with the pairwise

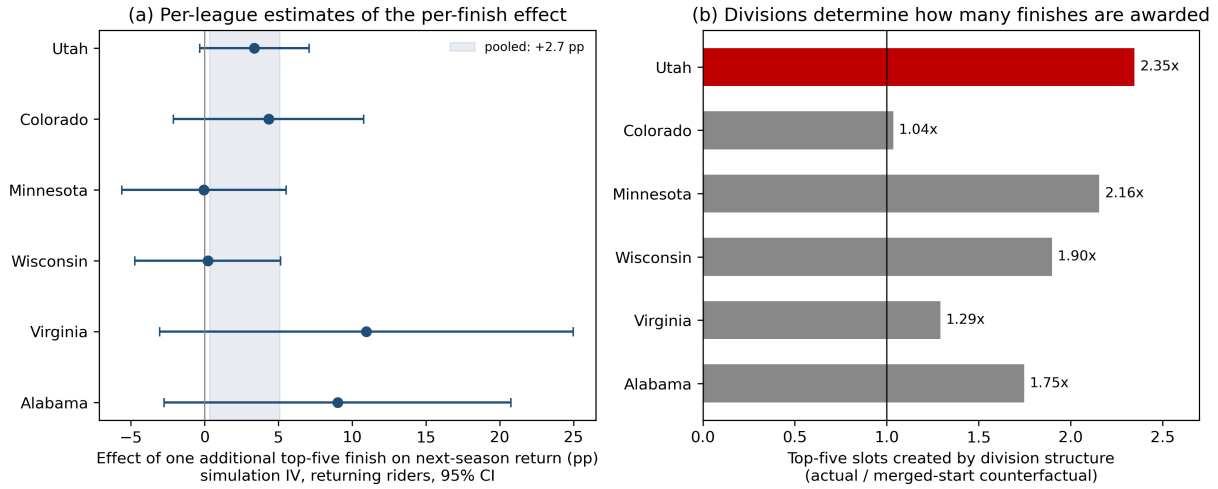


Figure 10: One price, different supply

Notes: Left: simulation-IV estimate of the per-top-five effect on next-season return per league (returning riders, 95 percent confidence intervals, rider-clustered); shaded band is the pooled estimate's 95 percent confidence interval, which is itself dominated by Utah's weight, so the Utah point's agreement with the band is partly mechanical. Right: ratio of actual top-five slots to slots under a merge-all-divisions counterfactual, holding attendance fixed.

Table 9: Division structure as recognition supply

League	Top-five finishes per 100 rider-seasons		Structure multiplier	Implied return-rate contribution (pp)
	Actual (with divisions)	Merged (one start per race)		
Utah	36	15	2.35×	0.6–1.1
Colorado	32	31	1.04×	0.0–0.1
Minnesota	34	16	2.16×	0.5–1.0
Wisconsin	42	22	1.90×	0.5–1.1
Virginia	57	44	1.29×	0.3–0.7
Alabama	53	30	1.75×	0.6–1.2

Notes: Partial-equilibrium accounting, not a behavioral prediction. The counterfactual merges each race's category-gender field into one start, attendance held fixed; championship races excluded; rates computed over each league's full panel of ranked, non-championship starts. The merged column awards exactly five slots per merged start. The implied contribution prices each league's difference in slots at the pooled per-finish range across the two pooled IV designs (simulation +2.7, opponent-strength +5.4 pp) and assumes the price is invariant to the merge; the within-paper dilution estimate implies this overstates the structural gain (see text), and non-Utah leagues are priced at a pooled estimate that Utah largely carries.

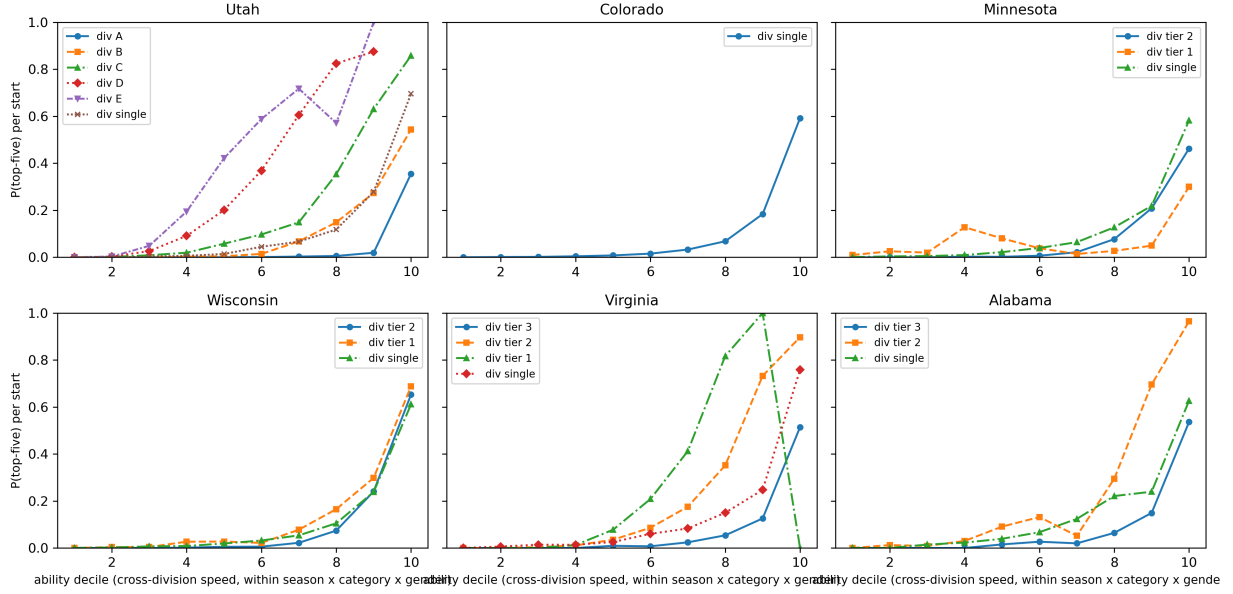


Figure 11: The incidence of recognition supply: top-five probability by ability and division

Notes: Probability of a top-five finish per start by ability decile (cross-division speed within season $\times$ category $\times$ gender) and division, per league. Where divisions exist, the curves separate: riders of identical measured ability face sharply different odds depending on their division, and mid-ability riders in lower divisions reach probabilities that mid-ability riders in merged or single-start fields (Colorado) reach in under 1 percent of starts.

model, and map the changes to persistence using the causal estimates. Respecting the unresolved mechanism, I do this under three mappings: a *recognition-count* mapping (changes in expected top-five counts, priced at the count estimate), a *standing* mapping (changes in expected fraction-beaten, priced at the standing estimate), and a *joint* mapping using the horse-race coefficients. Table 10 reports all three side by side.

Three lessons. First, under the count and joint mappings, recognition-broadening designs raise persistence: fixing field size at 25 raises the predicted return rate by 0.8 percentage points and triples the medal supply; top-ten recognition raises it by 0.4; collapsing to a single group lowers it by about 0.3. These are modest annual numbers—their significance is cumulative, as the next section shows. Second, under the pure standing mapping every aggregate effect collapses toward zero (no policy moves the return rate by more than about 0.2 percentage points), exactly as the framework predicts: standing is zero-sum, so design can only redistribute it. The aggregate case for recognition-broadening design rests on the

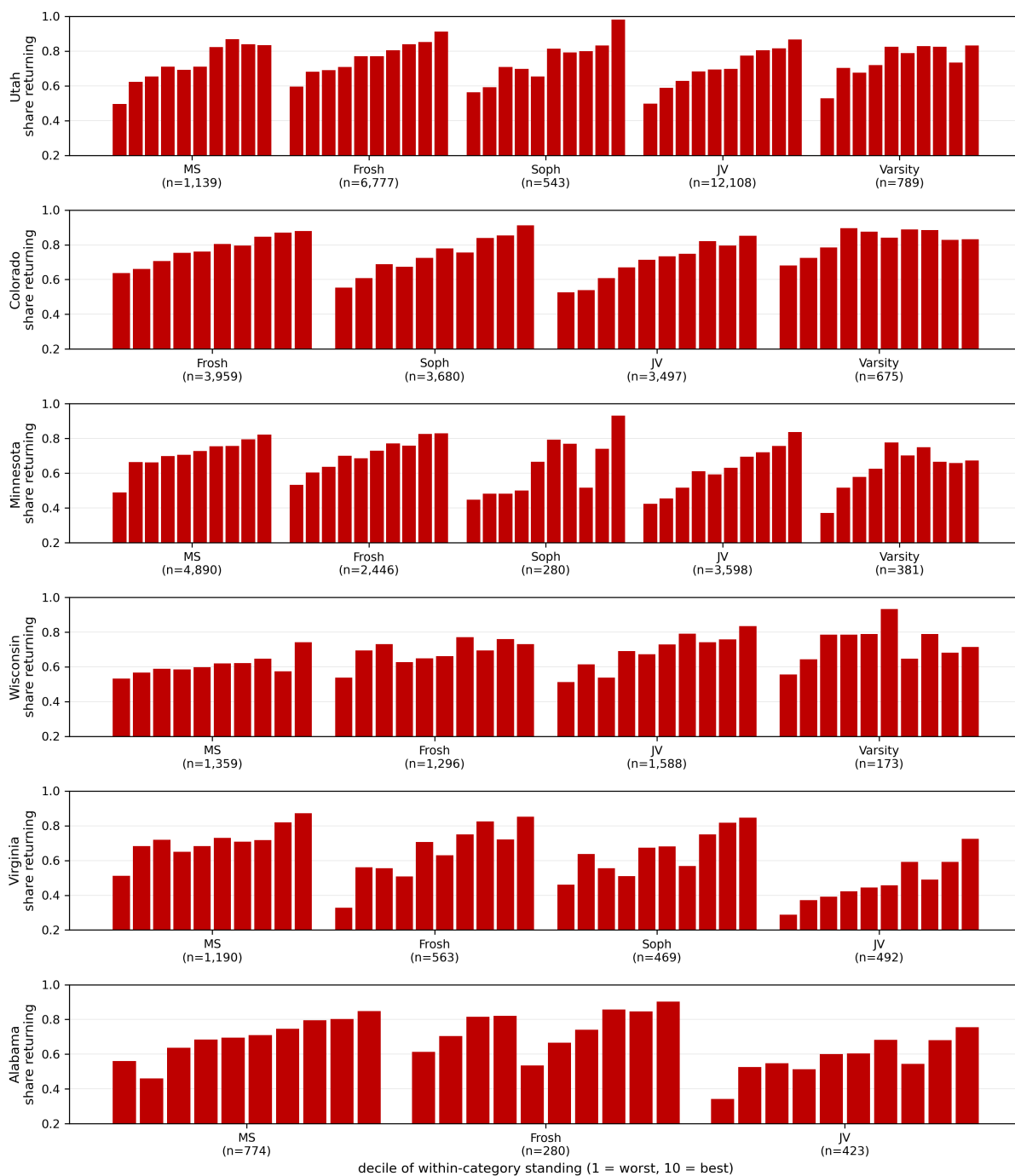


Figure 12: The persistence-by-rank gradient in all six leagues

Notes: Next-season return rate by within-category finish decile, by league and category (Middle School, Freshman, Sophomore, JV, Varsity); grade-12 rider-seasons excluded throughout. Top-minus-bottom-decile gaps range from 15 to 52 percentage points and appear in every league, including untracked Colorado—the descriptive gradient conflates skill sorting with causal effects of success, motivating the instrumental-variable designs.

Table 10: Annual policy counterfactuals under three mechanism mappings

Policy	Predicted return rate (%)			Medals per 100	Top-decile talent retained (joint)
	Count	Standing	Joint		
Current structure	65.2	65.2	65.2	22.3	2,513
One single group	64.9	65.1	64.7	12.7	2,490
One fewer division	65.0	65.1	64.9	17.6	2,495
One more division	65.4	65.0	65.5	34.9	2,517
Top-3 recognition	64.9	65.1	64.7	13.9	2,460
Top-10 recognition	65.6	65.1	65.8	45.8	2,546
Fixed field size 25	66.0	65.0	66.3	68.1	2,525

Notes: Each season’s rider pool held fixed; policies re-assign riders and recompute expected recognition and standing with the pairwise model, mapped to persistence with the count estimate, the standing estimate, or the joint (horse-race) coefficients. “Top-decile talent retained” counts predicted returners above a fixed top-decile ability threshold (anchored to the current-policy ability distribution, pooled across seasons), under the joint mapping. The dilution estimate (Section 7.2) implies the count-mapping gains for high-medal-share policies (top-10 recognition, fixed fields of 25) are upper bounds.

count mechanism being at least partly operative; the distributional case (moving persistence from riders who would stay anyway to marginal ones) survives either way. Third, the retention–selection frontier is nearly flat: across policies the mean ability of predicted returners barely moves, and the top-decile talent retained varies by roughly one percent. Inclusive recognition buys participation at essentially no cost in the average quality of the retained pool (Figure 13a); the gains accrue disproportionately to middle-ability riders, the group most marginal to the participation decision (Figure 13b).

Two further pricings discipline these numbers, and both are now computed rather than conjectured. Priced at the external design’s estimate (+1.6 per finish), the annual gains shrink to +0.14 percentage points for top-ten recognition and +0.26 for fixed-25 fields; priced with the estimated dilution (a share-dependent per-finish effect, floored at zero), they are +0.29 and +0.63. The policy *ranking* is identical under all five mappings; the magnitudes span roughly a factor of four, and I treat the count-mapping column as the upper end of that range rather than the headline. All mappings price only the persistence response; entry responses (recognition-rich designs may also attract entrants) are outside the exercise and would push the other way.

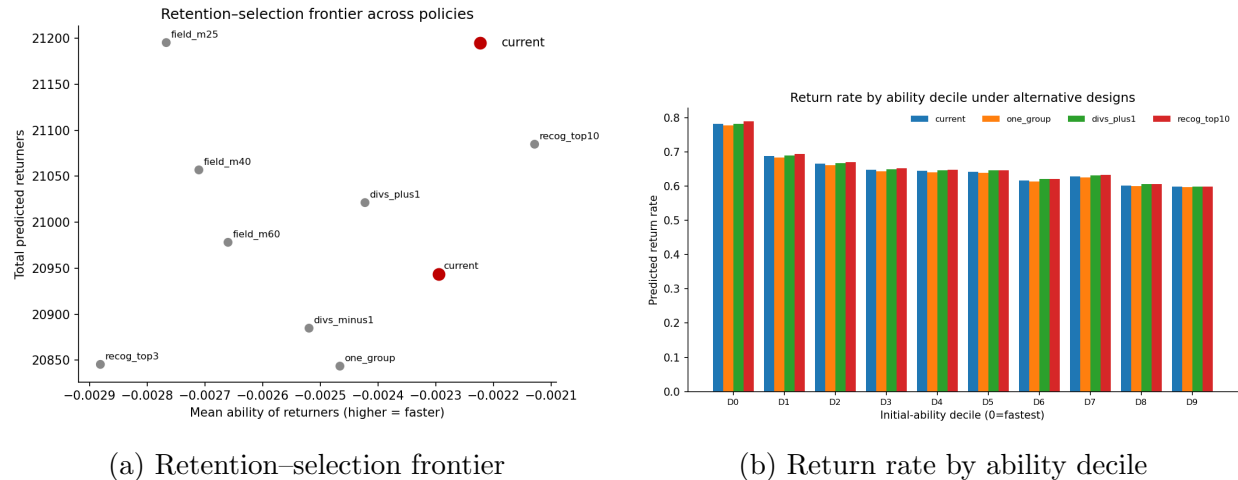


Figure 13: Broader recognition retains more riders at little cost in average ability.

10 Dynamic Microsimulation

The annual exercise holds the pool fixed and so misses the cumulative consequence of retention: a rider kept this season races again next season, accumulates development, and faces another retention draw—and exit is absorbing. I therefore build a dynamic microsimulation that follows the real entering cohorts forward under each design. Entry is exogenous (real entrants with their real first-season ability); the endogenous dynamics are division assignment (with the CPT ratchet under the current system), recognition from the pairwise model, persistence calibrated to observed return rates and the causal estimates, and ability evolution. I report the conservative scenario S1 (age effects and mean reversion only) throughout.²

Because the simulation inherits the mechanism question, its conclusions are stated under the count-based mapping and should be read as conditional on that mechanism; under a pure standing mapping the aggregate dynamic gains, like the annual ones, would largely vanish. Two further reconciliations bound the headline. First, the within-simulation *annual* return-rate gap under top-ten recognition is +8.0 to +8.2 percentage points per year—roughly

²Two errors in an earlier version of the simulation are corrected here. First, the category-placement ratchet was applied asymmetrically under recognition-depth policies, mechanically inflating their effects. Second, the persistence responses to recognition and standing were each calibrated from single-treatment regressions and then applied jointly, double-counting their common variation; the simulation now uses jointly estimated (horse-race) response coefficients. Both corrections reduce the policy effects relative to earlier drafts.

twenty times the annual counterfactual’s +0.4—because the simulation’s synthetic divisions (target size 30) sit in a far higher medal-share regime than the observed fields; the dynamic headline therefore embeds the simulation’s field structure, not new compounding magic, and inherits the dilution caveat of Section 7.2 with full force. Second, the heterogeneity table leaves the first-year recognition response genuinely uncertain (the simulation design says +2.2, the external design says zero, and both sit far below the returning-rider estimates). Re-running the simulation with the entrant response set to zero and the returning-rider response at a conservative +3.8 (between the two designs’ returning-rider estimates) cuts the senior-rate gain to +2.3 percentage points and the varsity-production gain to +11 percent. The defensible statement is therefore a range: top-ten recognition raises the senior-reach rate by between roughly 2 and 11 percentage points, the lower end if entrants do not respond to recognition at all. With that conditionality explicit: under the baseline response, the share of entrants who reach their senior year rises from 30.4 to 41.8 percent (+11.3 percentage points), and the production of varsity-level seniors rises by roughly a third. Decomposing the varsity-level gain (Figure 14a), about nine-tenths flows through the participation channel—more riders surviving to be varsity-eligible—and the small remainder through ability. Consistent with the tier-split reduced form (Table 3), the extra varsity seniors come disproportionately from middle-ability entrants (Figure 14c): the design lever broadens the top track’s membership; it does not manufacture top-decile ability, and I do not claim otherwise.

10.1 Bridging the annual and dynamic magnitudes

A +0.7 percentage-point annual return-rate gain becoming an +11.3 point senior-rate gain demands an explanation, not an assertion. Table 11 walks the gap in steps. Level L0 applies the annual gain analytically—compounding the static return-rate change over a career with no field feedback—and yields only +0.7 points. Level L1 turns on survival compounding within the simulation’s endogenous fields, holding every rider’s ability frozen: +10.7 points.

Level L3 adds ability evolution (full S1): +11.3 points. The amplification therefore comes almost entirely from L1—retained riders re-enter next season’s fields, where the policy again raises their recognition and survival, and the smaller simulated divisions under recognition-rich policies raise per-rider recognition rates above what the static calculation assumes—and almost none from ability dynamics. Propagating the uncertainty in the causal parameter through the simulation (draws of the persistence response) puts the senior-rate gain in a range of 10.1 to 11.5 percentage points. The full-simulation replication mean is +11.1; the bridge configuration’s +11.3 sits inside the draw range.

Table 11: Bridging the annual estimate to the dynamic simulation (top-10 recognition, S1)

Level	Senior-reach rate (%)		
	Current	Top-10 recognition	Gain (pp)
L0: Analytic compounding of the annual gain	17.9	18.6	+0.7
L1: Survival compounding in simulated fields (abilities frozen)	28.4	39.2	+10.7
L3: Full simulation, S1 ability evolution	30.4	41.8	+11.3

Notes: L0 compounds the annual (cross-sectional) return-rate gain over a career with no feedback. L1 simulates careers with retention feeding back into field composition and recognition, with abilities frozen at entry values. L3 is the full S1 simulation. Varsity-level production rises roughly 40 percent at both L1 and L3, confirming the amplification source is survival compounding in the simulation’s endogenous fields, not ability evolution. Full-simulation replication mean gain: +11.1 pp; range across causal-parameter draws: 10.1–11.5 pp.

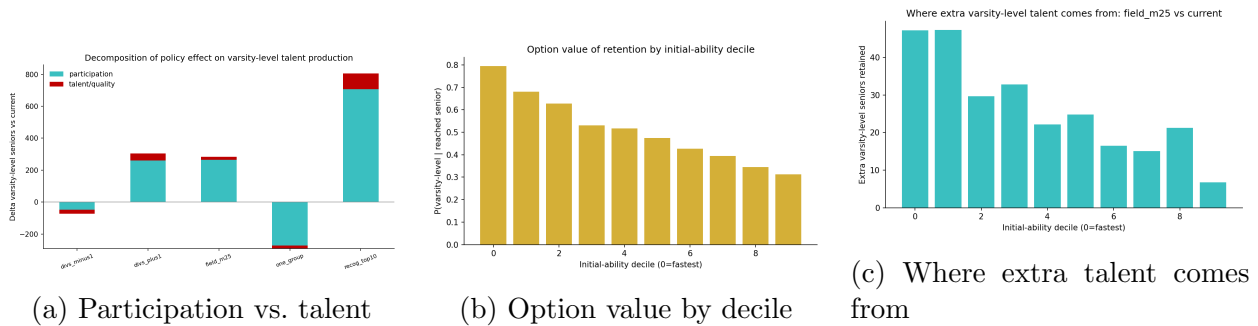


Figure 14: The varsity-level talent gain from broader recognition is a participation effect concentrated among middle-ability entrants.

10.2 Validation and limitations

The simulation is calibrated to reproduce observed return rates by grade, gender, and ability, and validated against the observed senior-participation profile (Figure 15). Two limitations are stated plainly. First, the simulation under-captures the steepness of the observed ability gradient in senior participation because it omits persistent unobserved commitment heterogeneity; the policy conclusions are relative comparisons across designs and are independently corroborated by the reduced forms of Section 5 and the model-light annual counterfactuals, but the levels should not be quoted as forecasts. Second, entry is held fixed: recognition-rich designs plausibly also *attract* entrants, a margin the simulation excludes, making the pipeline gains conservative on that dimension—while the dilution evidence cuts the other way, as in the annual exercise.

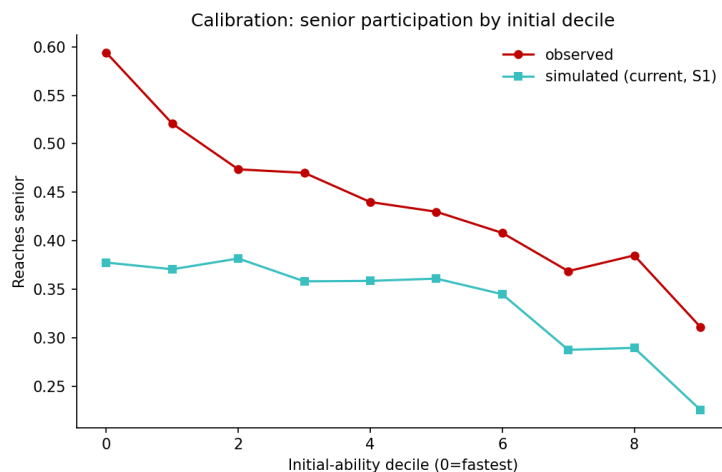


Figure 15: Simulation validation: senior participation by initial decile

Notes: Observed vs. simulated (current policy, scenario S1) senior participation. The simulation reproduces the level but under-captures the gradient.

11 Conclusion

Ability grouping is usually evaluated as a pedagogy. This paper evaluates it as a retention technology. In a setting where the instruction channel of tracking is absent by construction

and the participation margin is a real annual choice, the local competitive success generated by group assignment causally raises continued participation: +5.0 percentage points per top-five finish in the primary design (+1.4 in the cleanest-exclusion design; the designs agree for returning riders, and the cascade design bounds effects within $[-16, +4]$). Because early performance only weakly predicts who eventually populates the top track—about one-third of varsity seniors were not early standouts—retaining marginal participants has option value, and the reduced forms show early rank luck moves ordinary entrants onto the varsity track without manufacturing elite ability. Under a recognition-count mechanism, design levers that broaden recognition (more and smaller groups, deeper honors) raise both participation and, compounding over careers, the supply of varsity-level athletes by economically meaningful amounts; under a pure standing mechanism those aggregate gains vanish and only the distributional effects of regrouping remain. The mechanism evidence—a horse race favoring counts, a residual conditional-standing effect, a precise null at the single-race medal moment, and significant recognition dilution—is not yet decisive, and I have deliberately left the paper agnostic, reporting every policy under both mappings rather than letting a preferred story pick the table.

Nor is this a story about one state. Across six leagues processed through one pipeline, the per-finish price is not statistically distinguishable across leagues—precisely measured in Utah, underpowered elsewhere—and what differs is structure: leagues that subdivide their fields mint roughly twice the recognition that their own merged fields would, a partial-equilibrium accounting worth roughly half a point to 1.2 points of aggregate annual persistence at the estimated price, while Colorado, which runs single starts, forgoes that margin entirely. Division structure, in other words, is a recognition-supply policy, and leagues already set it very differently.

For education policy, the analogy is direct but must be labeled what it is: speculative extrapolation. The levers studied here—how many ability groups to form, how large to make them, how deep recognition should reach—map onto honor rolls and dean’s lists (how

deep should named recognition go?), course leveling (how many tracks, with rank visible within each?), and gifted-program size (a small elite pond versus broader designation). The findings suggest the participation-relevant content of such structures is the recognition and local standing they generate rather than the labels they confer, that recognition depth is a real but dilutable resource, and that in elective activities—where students can simply not come back—the option value of keeping middling performers engaged is large because early signals are poor. Whether those lessons transfer to settings where stakes, durations, and selection differ is exactly the kind of question the cross-context comparison of Appendix F shows cannot be settled casually.

What remains unresolved: the simulation-versus-external gap among first-year riders is unresolved; the reported standard errors exclude generated-regressor uncertainty pending a full-pipeline bootstrap; the dynamic projections are conditional on the count mechanism and on calibration choices that richer commitment data could sharpen; and the untracked-benchmark comparison did not deliver the compression pattern a simple tracking story predicts. None of these caveats threatens the core finding—that locally generated competitive success, quasi-randomly assigned, keeps young athletes in the pipeline—but they bound how far the design arithmetic built on it should be trusted.

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Appendix A Rider Identification and Linkage Audit

Procedure. Rider identifiers are constructed in stages. Names are cleaned and standardized; each rider’s grade is inferred from bracket names (e.g., seventh- versus eighth-grade middle-school brackets, Freshman = grade 9) and team fields; an expected graduation year is computed from the grade signals; and same-name riders are split when their records imply conflicting graduation years, different schools, or impossible career-timeline gaps. The procedure resolves 17,729 riders from 148,876 race-result rows. Grade-12 rider-seasons are excluded from all persistence regressions because seniors mechanically cannot return.

Audit. Because a missed link is observationally a dropout, I audit the linkage directly (Table A1). The resolved identifiers contain zero graduation-year conflicts; 2.1 percent of riders are associated with more than two schools (plausible transfers). For each apparent dropout (a non-senior rider who disappears), I search subsequent seasons for near-miss records—same school and gender with a similar name—that could be the same rider under a failed match; 1.2 percent of apparent dropouts have such a candidate, an upper bound on false dropouts from linkage failure. The headline return rate moves from 71.9 percent under loose (name-and-gender) matching to 71.7 percent under the baseline procedure—a 0.2-point difference—and to 64.2 percent under strict (name-gender-school) matching, which mechanically breaks school-switchers’ links. Finally, linkage difficulty is balanced with respect to the identifying variation: an indicator for having a common surname (2.8 percent of rider-seasons) is unrelated to the simulation instrument ($t = 0.1$), so match quality does not covary with the variation that identifies the causal estimates. Out-of-state moves, injuries, and switches to other sports remain observationally equivalent to quitting.

Table A1: Linkage audit

Metric	Value
Riders resolved	17,729
Share of rider IDs with conflicting graduation years	0.0%
Share of rider IDs with more than two schools	2.1%
Apparent dropouts (non-senior)	7,664
Near-miss candidate rate among dropouts (upper bound on false dropouts)	1.2%
Return rate, loose matching (name + gender)	71.9%
Return rate, baseline (v2 identifiers)	71.7%
Return rate, strict matching (name + gender + school)	64.2%
Common-surname share of rider-seasons	2.8%
Common-surname balance on simulation instrument (t)	0.1

Notes: Strict matching mechanically codes school-switchers as dropouts, so its lower return rate is expected. The loose-versus-baseline gap of 0.2 pp bounds the sensitivity of the outcome to over-splitting.

Appendix B Pre-Treatment Skill

Skill is the mean residual from a regression of log average lap time on venue, year, category, and laps ($R^2 = 0.41$), measured using only pre-treatment (prior-season) information and shrunk toward grade $\times$ gender $\times$ year cell means by empirical Bayes. The measure is highly stable (season-to-season autocorrelation 0.82). The underlying season-mean residual is itself reliable rather than noise-driven: its split-half correlation across a rider’s odd- and even-numbered races within a season is 0.74, and its year-over-year correlation among returning riders is 0.60.

Entrant timing. First-season entrants have no prior season, so an entrant ability measure built from the full first season would embed the same races that generate first-season recognition—post-treatment variation. All long-run analyses therefore use a race-one-only construction for entrants: ability and expected recognition are computed from the first race alone, with outcomes defined over subsequent races. Table A2 compares the long-run reduced forms across constructions. The varsity-margin results are robust—+32.0 per unit of race-one top-five rate ($t = 7.2$), +14.8 per race-one expected count ($t = 8.6$), +9.9 in the

original (full-first-season) variant ($t = 12.1$)—while the cumulative-seasons effect, significant in the original variant, is a null under the strict timing, and I treat it as such in the text.

Table A2: Long-run reduced forms under alternative entrant-timing constructions

Outcome	Race-1 rate		Race-1 count		Original (full season)	
	Coef.	t	Coef.	t	Coef.	t
Future seasons (count)	+0.06	0.5	−0.01	−0.3	+0.13	6.6
Ever senior (pp)	+7.3	1.6	+3.1	1.7	+4.7	5.8
Ever varsity (pp)	+32.0	7.2	+14.8	8.6	+9.9	12.1
Top talent (pp)	+30.7	9.0	+12.2	8.6	+2.8	6.0
Varsity/elite senior (pp)	+21.4	6.2	+10.1	7.1	+5.8	10.2

Notes: Specifications as in Table 2. $N = 8,458$ for the race-one variants, 9,249 for the original variant. Coefficient units differ across variants (per unit rate, per expected count); signs and significance are the comparable objects.

Split-sample construction. Because ability is an estimated, shrunken regressor entering both the control and the instruments, Appendix Table A10 includes a split-sample specification in which ability is estimated from odd-numbered races and the instrument is built on even-numbered races, eliminating correlated estimation error by construction; the estimate is +4.1 ($t = 6.4$), attenuated as expected under classical measurement error. Results are also robust to using unshrunk ability.

Appendix C The Expected-Rank Engine

The pairwise model treats course-adjusted race time as ability plus a mean-zero shock with SD σ ; rider j beats i with probability $\Phi((A_i - A_j)/(\sigma\sqrt{2}))$. Expected rank and fraction-beaten are computed in closed form; win and top- k probabilities are computed by Monte-Carlo ranking so that exactly k riders are top- k in each draw. The Monte-Carlo draws resample *empirical* race-day residuals rather than normal deviates, so the fat tail of race-day disasters (mechanicals, crashes) is reflected in the top- k tail probabilities by construction. Figure A1 reports calibration: Spearman correlation of predicted and realized rank 0.74;

top-five AUC 0.87.

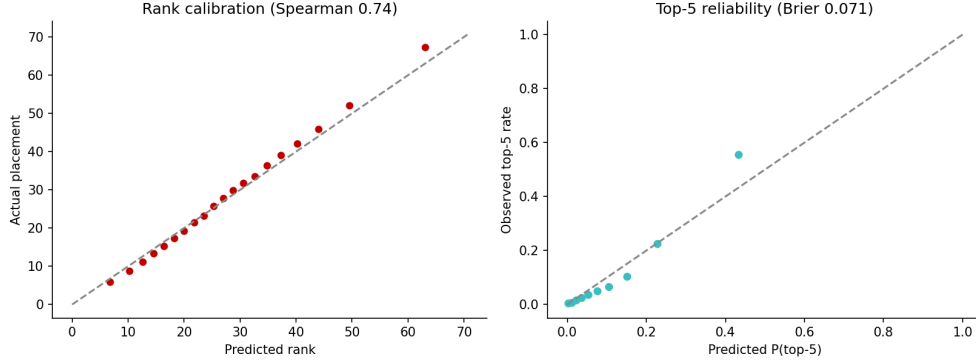


Figure A1: Predicted vs. actual rank and top-five reliability

Appendix D The Cascade Design: Density, Inference, and Compliers

Figure A2 shows the density of the CPT running variable; the absence of bunching at division cutoffs is consistent with the identifying assumption (riders and coaches do not precisely manipulate CPT rank around cutoffs). Table A3 collects the weak-instrument-robust inference: the 95 percent Anderson–Rubin set for the top-five count effect is $[-16, +4]$ percentage points at baseline and $[-32, +9]$ in a donut specification excluding riders within two positions of a boundary; randomization inference re-drawing cutoff placements 200 times yields $p = 0.46$ for the reduced form (-1.3 pp). A stacked-boundary design (riders within 2–8 CPT positions of a cutoff, boundary fixed effects, distance trends) finds no first stage at the predicted discontinuities ($t \in [-0.1, 1.0]$), consistent with the rule reproducing only 42 percent of assignments near cutoffs. The design’s correlation with predetermined commitment is small (prior racing: -0.013 , $t = -2.6$; Table A7).

Table A4 reports the slot-ladder promotion design built from the placement table’s own logic (Section 6): returning riders default to their period- T division and vacated seats are filled in period- T points order, giving each returner below the top division a predicted promo-

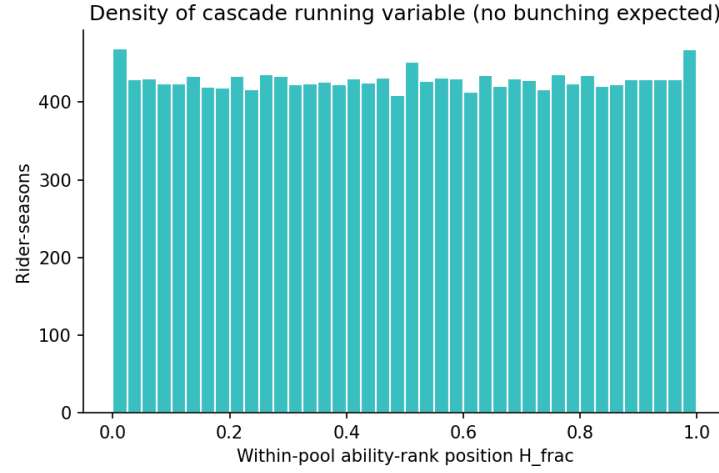


Figure A2: Density of the cascade running variable

Table A3: Cascade inference

Specification	AR 95% set (pp)	N
Baseline	$[-16.0, +4.0]$	17,164
Donut (≥ 2 positions from boundary)	$[-32.0, +9.0]$	16,126
Randomization inference (200 permutations of cutoffs)	RF = -1.3 pp, $p = 0.46$	

Notes: Anderson–Rubin sets for the effect of one top-five finish on next-season participation under the cascade design.

tion cutoff. The first stage is stable across bandwidths; the persistence effect at the margin is a bounded zero for the power reasons discussed in the text.

Table A4: The slot-ladder promotion design

	± 5 ranks	± 8 ranks	± 12 ranks	all (quadratic)
First stage: actually promoted	+0.12 ($t = 2.2$)	+0.11 ($t = 2.4$)	+0.14 ($t = 2.6$)	+0.07 ($t = 2.1$)
$T+1$ local standing (frac. beaten)	-0.03 ($t = -1.0$)	-0.03 ($t = -1.1$)	-0.05 ($t = -1.4$)	+0.02 ($t = 0.6$)
Persistence (raced $T+2$, pp)	-7.2 (8.2) ^a	+2.3 (8.2)	-1.4 (10.0)	+5.0 (4.0)
N (riders near cutoff)	924	1,428	2,030	3,669
Validation: ratchet violations 2.3%; promotion-points-rank Spearman -0.64 (right-signed in 100% of cells); capacity absorption -0.46 (vs. 1.0 = full absorption); 96 cutoff cells.				

Notes: Cell (pool $\times$ year $\times$ source-division) fixed effects with side-specific linear distance trends (± 12 : quadratic); standard errors clustered by pool-year. Persistence rows use the non-senior, non-censored subsample (smaller N).

^a Persistence point estimates are noisy across bandwidths; the design is power-limited, not informative, at this margin (see text).

Table A5: Supporting estimates referenced in the text

	Coef.	(s.e.)	t	N
<i>Panel A. Graduation-slack reduced form (exposed riders, margins B+C stacked)</i>				
Senior share above $\rightarrow$ raced next season	+0.87	(0.32)	2.7	10,555
balance: exposed riders' ability	-0.02	—	-0.2	12,942
balance: share first-year	+1.17	—	1.8	12,942
<i>Panel B. Recognition dilution (2SLS, persistence on top-five count)</i>				
Top-five count	+0.050	(0.006)	7.8	28,434
Top-five $\times$ medal share (demeaned)	-0.188	(0.046)	-4.1	28,434
<i>Panel C. Cascade-eligible subsample (returning riders, division brackets)</i>				
Simulation design: top-five count	+0.044	(0.009)	5.0	14,958
External design: top-five count	+0.009	(0.011)	0.8	14,958

Notes: Panel A: reduced form of next-season participation on the senior share of last year's upper-division riders, for riders not protected at the margin; pool and year-by-margin fixed effects, lag size, pool sizes, ability deciles, and experience controls; clustered by pool-year. Balance rows regress the listed characteristic on the instrument. Panel B: 2SLS as in Table 4 with the medal-share interaction instrumented by the instrument-share interaction. Panel C: the main specifications re-estimated on the cascade-eligible subsample.

Table A6 characterizes compliers via subgroup first stages. The simulation and external instruments move recognition broadly (first stages of 0.5 to 1.3 across subgroups). The cascade's first stage is concentrated among returning riders (0.29) and the fast initial-ability tercile (0.56), and is weak for girls (0.14) and the slow tercile (0.08): its LATE is for fast,

returning riders near division-size cutoffs—the population on which all three designs agree (Section 7). No subgroup shows a sign-reversed first stage for any design, the testable implication of monotonicity.

Table A6: Subgroup first stages (complier characterization)

Subgroup	Simulation		External		Cascade	
	Coef.	t	Coef.	t	Coef.	t
All	0.71	50.2	1.07	44.3	0.27	7.9
Boys	0.68	41.3	1.08	34.7	0.31	7.8
Girls	0.70	23.9	1.03	24.3	0.14	2.0
First-year	0.67	43.2	1.31	40.6	0.26	3.7
Returning	0.74	29.7	0.89	26.1	0.29	7.5
Fast tercile	0.77	44.8	1.09	34.7	0.56	9.0
Slow tercile	0.51	9.8	0.68	11.6	0.08	2.4

Notes: First-stage coefficients of realized top-five count on each instrument, by subsample, with controls as in Table 4. Cascade subsamples restricted to cascade-eligible rider-seasons.

Appendix E Robustness Appendix

This appendix collects, as numbered tables, every robustness claim made in the text.

E.1 Balance and pre-trends

Table A7 regresses predetermined variables on each instrument, conditional on the baseline controls. The first panel—prior-season participation regressed on the season- s instruments—is the pre-trends placebo: standing-type instruments are balanced ($t \approx -0.9$); count-type instruments show small residual correlations (e.g., -0.015 , $t = -6.5$ for the simulation count instrument), precisely estimated but an order of magnitude too small, and of the wrong sign, to generate the estimated effects. Lagged performance panels load on all instruments conditional on ability deciles, as functions of lagged ability must; the operative check is the stability of the estimates under much richer $f(A)$ (Table A10). Table A8 isolates the

experience exhibit: without experience controls the standing instrument is wildly imbalanced on prior participation (-0.51 , $t = -34.8$); with them the imbalance vanishes (-0.004 , $t = -1.0$).

Table A7: Balance of predetermined variables across instruments (conditional on controls)

Predetermined variable	Instrument	Coef.	t	N
Raced previous season	Simulation top-5 (std)	-0.010	-5.8	32,131
	Simulation fraction-beaten	-0.004	-1.0	32,131
	External top-5 (std)	-0.013	-5.0	32,131
	External fraction-beaten	-0.008	-0.9	32,131
	Cascade top-5	-0.010	-2.2	17,164
Lagged top-5 count	Simulation top-5 (std)	0.302	12.5	16,301
	External top-5 (std)	0.451	13.1	16,301
	Cascade top-5	0.379	9.1	11,809
Lagged fraction beaten	Simulation top-5 (std)	0.007	1.6	16,301
	External top-5 (std)	0.012	1.8	16,301
	Cascade top-5	0.043	4.6	11,809
Lagged races entered	Simulation top-5 (std)	-0.051	-4.0	16,301
	External top-5 (std)	-0.146	-8.9	16,301
	Cascade top-5	-0.063	-2.7	11,809

Notes: Each row regresses the predetermined variable on the instrument, conditional on the baseline control set (ability deciles, experience controls, region $\times$ year FE). The first panel (prior-season participation on current instruments) is the pre-trends placebo. Lagged-performance correlations are expected: the instruments are functions of lagged ability; see text. Rider-clustered t -statistics.

Table A8: The experience confounder: balance of prior participation on the standing instrument

Controls	Coef.	t	N
Skill and year fixed effects only	-0.509	-34.8	32,131
+ experience controls (races entered, first-year status)	-0.004	-1.0	32,131

Notes: Prior-season participation regressed on the simulation fraction-beaten instrument. A fast newcomer mechanically dominates an easy field, so expected standing is correlated with being new—which independently predicts attrition—until experience is controlled.

E.2 Attendance purge

Table A9 reports the headline estimates under the old (raw seasonal count) and new (per-race rate $\times$ median three-race season) treatment definitions, plus a fully rate-on-rate specification.

Table A9: Purging attendance from the count treatment

Specification	2SLS	(rider SE) [field SE]	First-stage F	N
Simulation: raw count (old)	+3.7	(0.5) [0.8]	3,223	28,434
Simulation: standardized (new)	+5.0	(0.6) [0.9]	2,577	28,434
Simulation: rate-on-rate	+15.5	(1.9) [2.7]	3,050	28,434
External: raw count (old)	+1.6	(0.6) [0.9]	2,474	28,434
External: standardized (new)	+1.4	(0.7) [1.0]	1,987	28,434
External: rate-on-rate	+4.5	(2.2) [3.1]	2,405	28,434

Notes: Outcome is next-season participation, pp. “Raw count” sums top-five finishes over attended races (attendance embedded); “standardized” is the per-race rate scaled to a median three-race season; “rate-on-rate” instruments the per-race top-five rate with the expected per-race rate (coefficient per 0 $\rightarrow$ 1 rate change). Races entered controlled in all rows. Controls as in Table 4.

E.3 Ability-control and fixed-effect ladders

Tables A10 and A11 report the estimate under progressively richer ability controls and fixed effects.

Table A10: Robustness to the ability control $f(A)$

Specification	2SLS	(rider SE) [field SE]	First-stage F	N
Simulation: ability-decile FE (base)	+5.0	(0.6) [0.9]	2,577	28,434
Simulation: 20 percentile bins	+5.0	(0.6) [0.9]	2,507	28,434
Simulation: $A \times \text{grade} \times \text{gender}$	+5.4	(0.7) [1.0]	2,350	28,434
Simulation: split-sample (odd/even races)	+4.0	(0.6) [0.8]	2,183	28,434
External: ability-decile FE (base)	+1.4	(0.7) [1.0]	1,987	28,434
External: 20 percentile bins	+1.1	(0.7) [1.0]	1,901	28,434
External: $A \times \text{grade} \times \text{gender}$	+1.4	(0.7) [1.0]	1,897	28,434

Notes: Top-five count effect on next-season participation, pp. The split-sample row estimates ability from odd-numbered races and builds the instrument on even-numbered races, eliminating correlated estimation error; attenuation is expected under classical measurement error.

Table A11: Robustness to fixed effects

Specification	2SLS	(rider SE) [field SE]	First-stage F	N
Simulation: year FE only	+4.9	(0.6) [0.9]	2,587	28,434
Simulation: region $\times$ year (base)	+5.0	(0.6) [0.9]	2,577	28,434
Simulation: school $\times$ year (absorbed)	+5.0	(0.7) [0.9]	2,264	27,900
External: year FE only	+1.5	(0.7) [1.0]	1,997	28,434
External: region $\times$ year (base)	+1.4	(0.7) [1.0]	1,987	28,434
External: school $\times$ year (absorbed)	+1.4	(0.8) [1.0]	1,880	27,900

Notes: Top-five count effect on next-season participation, pp. The school $\times$ year specification compares riders on the same team in the same season. Singleton school-years are dropped ($N = 27,900$).

E.4 Speed outcomes: no performance response

Table A12 supports the participation-not-performance discussion in Section 7.2.

Table A12: Recognition does not move measured speed

Specification	Coef.	(s.e.)	First-stage F	N
Next-season pace, baseline ability controls	+1.58	(0.29)	599	9,860
Next-season pace, $A \times \text{grade} \times \text{gender}$	+0.06	(0.34)	440	9,860
In-contention race pace (rider $\times$ season FE)	−0.93	(0.42)	—	65,647

Notes: Units: percent of lap speed (course-adjusted log average lap time $\times (-100)$; venue, season-year, and gender $\times$ grade benchmarks; penalty rows corrected to riding time). Rows 1–2: 2SLS effect of an additional top-five finish on next-season pace, external-opponent instrument, returning riders with at least two next-season races; the baseline-control row documents the ability-control artifact that the interacted controls eliminate. Row 3: race-level within-rider $\times$ season estimate of riding faster when the day’s start field puts the rider in top-five contention (expected rank ≤ 7 from prior-season skill); the full expected-rank profile is flat within ± 0.15 percent. Rider-clustered SEs. The full design space (alternative pace measures, censoring, selection bounds, promotion-cutoff RD) is archived with the replication materials.

Appendix F The Untracked Benchmark: A Descriptive Comparison to Alpine Racing

If ability grouping protects low-ranked participants from dropping out, a parallel sport *without* within-age ability tiers should display steeper persistence-by-rank gradients. Utah offers a natural candidate: Intermountain Division alpine ski racing, which serves the same ages in

the same state but groups athletes only by age and gender, with no ability divisions. I construct harmonized return-rate gradients—same outcome (raced the next season), same decile construction (within age-group $\times$ gender, pooled across birth years), both genders, raw and ability-adjusted—for alpine and for two MTB analogues: the within-division gradient (the tracked object) and the within-grade-gender gradient pooled across divisions (the gradient MTB would display if it were untracked).

The comparison, shown in Figure A3 and Table A13, does *not* deliver the clean contrast the tracking framing predicts, and I report it as descriptive context rather than evidence. The raw alpine gradients are large: a 24.9-percentage-point return-rate gap between top and bottom decile for boys and 20.0 for girls. But the harmonized MTB analogues are similar or larger (raw pooled: 32.5 boys, 20.1 girls; raw within-division: 29.6 and 20.9), and after ability adjustment the alpine gradients (14.7 and 5.6) are *smaller* than the MTB ones (within-division adjusted: 27.8 and 17.8). Taken at face value, riders’ persistence in this league is at least as rank-sensitive within their ability-matched divisions as alpine racers’ persistence is within untracked age groups—the opposite of a simple compression story. The honest reading is that cross-sport comparisons of this kind are confounded by everything that differs between the sports: cost structure (alpine is dramatically more expensive, which may itself flatten rank-sensitivity by selecting committed families), club versus school institutions, season structure, and the underlying populations. The within-league causal evidence and the same-sport cross-league comparison of Section 8, not this comparison, carry the paper’s claims; the alpine benchmark is included because the tracking framing predicts a pattern here, a referee would ask for it, and reporting that it does not cleanly appear is more informative than omitting it.

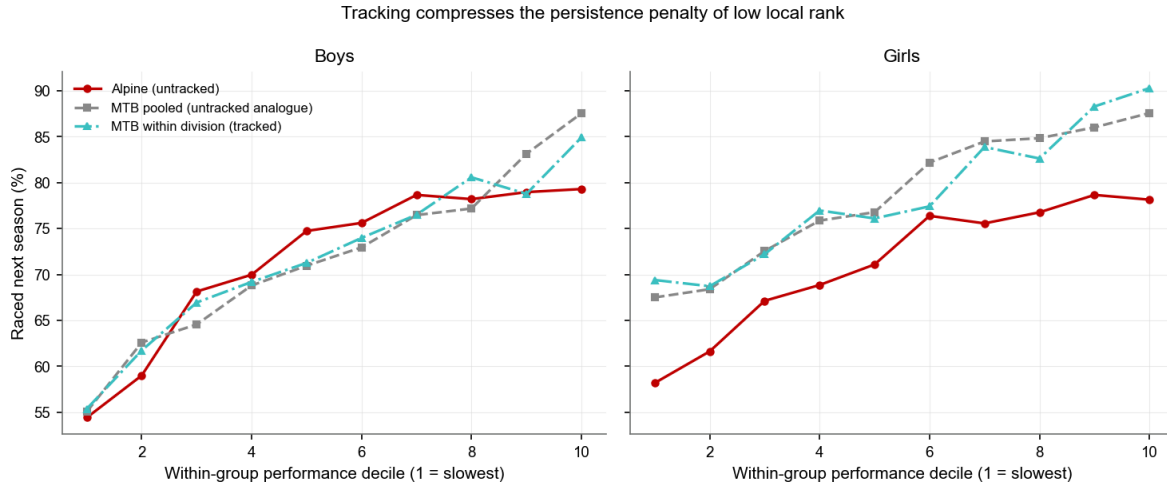


Figure A3: Return rate by performance decile: alpine (untracked) versus MTB

Notes: Harmonized return-by-decile gradients. Alpine: within age-group×gender deciles. MTB: within-division deciles (tracked) and within-grade-gender deciles pooled across divisions (untracked analogue). Raw and ability-adjusted variants; see Table A13.

Table A13: Top-minus-bottom decile return-rate gaps (pp), harmonized

Sport	Series	Boys	Girls
Alpine (untracked)	Raw	24.9	20.0
Alpine (untracked)	Adjusted	14.7	5.6
MTB, pooled grade-gender (untracked analogue)	Raw	32.5	20.1
MTB, pooled grade-gender (untracked analogue)	Adjusted	24.0	12.9
MTB, within division (tracked)	Raw	29.6	20.9
MTB, within division (tracked)	Adjusted	27.8	17.8

Notes: Gap in next-season return rate between the top and bottom performance decile within the indicated grouping. “Adjusted” nets out prior-season ability. The harmonized comparison does not show the tracked gradients to be compressed relative to the untracked sport; see text for caveats (different sports, cost structures, institutions, eras). Descriptive only.